IOT BASED FLOOD PREDICITION SYSTEM FOR AGRICULTURAL FIELDS USING ML

A PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF

BACHELOR OF TECHNOLOGY
IN
ELECTRONICS AND INSTRUMENTATION ENGINEERING

BY

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UNDER THE GUIDANCE OF

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ABSTRACT

Due to rapidly fluctuating climate trends and excessive rainfall, agriculture zones have become more prone to sudden floods that affect crops and other infrastructures. Villages do not have any automatic monitoring systems for providing early warnings for any floods. In this case, it takes time to alert farmers about floods, and by the time they get the warning, water level boundaries will have already been breached. Therefore, this research proposes an Internet of Things (IoT) based system for predicting floods in agriculture farms.

The suggested approach involves the deployment of an Arduino based sensor node integrated with water level, rain, and atmosphere sensors to continuously monitor the environment within the vicinity of the canals and drainage channels. The gathered data from sensors through the LoRa wireless connection is sent to the Raspberry Pi based gateway that collects the data and logs it into a CSV file, generates its graphical representation on the locally based web dashboard, and sends it to the remote-based IoT server named ThingSpeak for further analysis. A Random Forest based classification algorithm is built using the gathered sensor data and used to classify each data record as normal or possible flood situation. In case the flood is identified by the classifier, the Raspberry Pi sends SMS alert messages to subscribed people via the GSM modem, whereas the current status is updated on the LCD display, ThingSpeak graph, and Raspberry Pi dashboard.

The second phase of the project involved integrating the hardware components, developing the data logging and dashboard software, implementing the Random Forest algorithm, and testing the performance of the model under varying environmental conditions. It is clear from the outcomes that the IoT–ML framework can accurately measure field parameters and deliver effective flood warning services.

Keywords: Internet of Things (IoT), Flood Prediction, LoRa, Random Forest, ThingSpeak, GSM Alerts.

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NOMENCLATURE

IoT	Internet of Things
LoRa	Long Range Radio communication technology
ML	Machine Learning
RF	Random Forest classifier
GSM	Global System for Mobile communication
CSV	Comma Separated Values (data file format)
LCD	Liquid Crystal Display
API	Application Programming Interface(ThingSpeak Key)

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CHAPTER 1
INTRODUCTION

1.1 Background of the Problem

Floods one of the worst disasters taking place in the world today. Millions of people across the world face floods every year. Agriculture as the most important livelihood of the world depends heavily on environmental conditions and changes in the conditions may pose a threat to the crops. The number of floods happening in the world has seen an increase in the past few decades as there are some natural and manmade reasons like climate change, rapid urbanization, deforestation, poor environmental management practices, etc.

The drastic changes taking place due to climate change is one of the reasons why flood events are increasing. Due to global temperature increase the rainfall is taking place erratically with higher intensity for a longer period. In many areas it has been observed that heavy and prolonged rainfall leads to an excess flow in the river, canal and lake, which exceeds their capacity and spills into the agriculture and residential areas. Deforestation also is a major reason, because removal of trees reduce the absorption capacity of the land and that also results in excess flow.

The agricultural lands that are in the close proximity to a water body, like a river, canal, lake and dam, face heavy threat. Such lands are suitable for agriculture because they have rich soil and there is always availability of water for irrigation. When there is a heavy rainfall or sudden release of water from dam, then there is rapid submersion of the land. The crops that were already in the agricultural land are damaged within minutes or hours and this makes the farmers face tremendous losses. Paddy, fruits and vegetables are susceptible to the problem of water logging, where roots of the crop may not get oxygen and that would destroy the plant. Erosion of top soil also takes place in such a case.

The traditional methods for flood prediction and management are manual and are based on observation only. In many regions, we have to physically measure the level of water in nearby rivers or lakes by using a simple instrument and also observe the climate pattern and listen to weather reports from the meteorology department to give the alerts to the community. However, these methods are either not precise, efficient or up to the mark.

The primary problem with these traditional flood monitoring and prediction methods is that it is time consuming and requires huge human effort to obtain the data, analyze and communicate. Many a time the information may be provided to the community at the wrong time, so much that preventive measures cannot be taken and crops are lost or damage occurs to the property.

Another drawback of these methods is that it considers only one or two factors when predicting the flood and they are mainly water level and rainfall. Other factors like soil moisture, atmospheric pressure, temperature, humidity etc are very important to determine and predict the floods more accurately and they are considered with less or no attention in such methods. Hence, the prediction models are not very accurate and are less reliable.

Therefore there is a need to incorporate advanced technologies and implement them in the flood detection and management systems. The need for the development of smart agriculture is already discussed by many researchers. Smart agriculture can be understood as a type of agriculture that involves the use of technology in the form of various tools and methods to improve farming practices.

Among the several technologies in smart agriculture that can be used to solve this issue, Internet of Things and Machine learning are ones that are gaining attention by many of the researchers. With the help of IOT it is possible to place sensors that will take the values of factors like water level, soil moisture, temperature and humidity of the atmosphere etc at real-time.

However, after collecting these data we can then use Machine learning in order to predict the floods. By training the algorithms the required prediction can be obtained and accordingly alerts can be given to the farmers and also the government that have a look over the management aspects.

It would be a perfect and better solution to use IOT and ML together for the better flood prediction and management of crops in agriculture areas because: it would make the prediction real-time, increase the prediction accuracy by using the combination of many features that affect floods, would reduce the dependency on the labor, and the information will be readily available with faster reaction time.

In conclusion, floods are one of the worst disasters that can happen to a particular agriculture domain. Traditional methods of flood detection are not up to the mark. It is high time that smart techniques should be applied. Through the application of IOT and Machine learning, better flood detection systems can be developed to reduce agricultural losses and secure the agricultural domain.

1.2 The Necessity and Importance of the Research

The need to develop an intelligent flood prediction system is becoming increasingly prominent as traditional monitoring systems fall short of meeting the demands posed by dynamic and multifaceted environmental circumstances. Traditional flood monitoring relies solely on manual inspection and rudimentary meteorological information which is seldom capable of facilitating prompt and accurate flood prediction. Traditional monitoring system fails to simultaneously process numerous environmental parameters to make flood warnings timely and accurate. Environmental changes are escalating due to changing climate patterns, the human activities, thereby necessitating the development of advanced and intelligent monitoring systems.

The farming systems have become vulnerable due to extreme variation in the environment. Farmers are experiencing unanticipated rain, inconsistent monsoon and fluctuating water level, which can readily flood the low-lying agricultural fields or areas situated near water bodies. Even the slight delay in flood warnings would lead to enormous crop loss and economic setback to the farmers in the future.

Therefore, it is essential to develop an automated and quick flood warning system for farmers, to enable them to take precautionary measures beforehand such as draining excess water from the fields, strengthening irrigation channels, temporarily stopping the agricultural work etc. It also decreases the reliance on traditional flood monitoring system, which require humans intervention and often provide information timely and accurately to rural and distant areas.

With the advent of newer technologies like the Internet of Things (IoT), the intelligent flood prediction system can be materialized. It enables an interconnection of different sensors that monitors environmental data in real time such as water level, soil moisture, temperature, humidity, rainfall and transmit this data wirelessly to the data acquisition device. Since all parameters are monitored real time, even minute change in environment is promptly detected.

Along with the IoT, the machine learning algorithms are used to analyze the data and to estimate the upcoming flood probability based on the trained patterns from the historical data. By using these algorithms the prediction accuracy can be improved with increased time and it becomes more robust than traditional systems.

Cost-effectiveness is also a key factor of this research, as most of the farmers belong to the developing countries and have an access to limited budget and advanced technology. So the developed system is desired to be very affordable and adaptable for wide range deployment. Hence this low-cost solution ensures a maximum benefit to the farming community especially small and marginal farmers.

There is another key factor to be taken into consideration regarding limitations of communication between different nodes of the system. In most rural areas there is either very less internet or no connectivity, which makes the communication between remote sensors and the main station inefficient. Long range communication protocols such as LoRa are suggested in this research, to achieve better communication and reduce the reliance on the internet for communication, as it is optimized for low power and long range.

The role of the embedded systems is important in this research for managing and monitoring, so platforms such as Raspberry Pi is utilized for data processing, communication, storing and as a central point for acquiring the data from sensors. This helps the developed system to become more efficient with in edge processing.

This research endeavors to develop an efficient and sustainable flood detection system for farmers utilizing modern IoT technology and machine learning algorithms that will address the risks associated with floods for agriculture and can save considerable crop loss and financial losses. The research also aims to contribute towards "smart agriculture" and climate resilient agriculture.

So, the necessity and importance of this research is due to the evolving traditional flood monitoring system, increasing impact of the climate change on farming and lack of smart technologies in most of the underdeveloped and developing countries, and a promising step for implementing the IoT and machine learning based flood warning system in agriculture.

1.3 Problem Statement

Despite advanced flood prediction systems such as sophisticated modelling approaches and satellite based observation technologies, a huge disparity still exists regarding the creation of affordable, feasible, and portable flood prediction systems for the purposes of agricultural sectors. Many of these existing systems are meant to be implemented on a large scale, and often rely on the use of high-cost infrastructures that would not be useful in the hands of the marginal and small farmers. Therefore, there is a need to develop such a system that will be not only capable of predicting the flood accurately but it should be low cost and flexible in nature for the rural areas.

One major drawback among the present flood prediction models that have been reviewed is their dependence on a narrow spectrum of parameters in general that will focus on observing the water levels on the rivers, canals, reservoirs or water bodies. While it is true that water level is a vital indicator that would enable forecasting the flood event. It is still an incomplete factor since flood occurs because of numerous natural factors. Floods are very complicated natural events occurring with various parameter inputs, for instance soil moisture, intensity of rain fall, temperature, humidity, atmospheric pressure etc. If none of these factors are taken into consideration, the information that is derived from it might be partial and thus the overall prediction made may not be correct.

For example, when observing the soil moisture component, it indicates how much amount of water that the soil would be able to absorb before the runoff starts. So if the land is already saturated with water from earlier rains then even the minimal amount of rainfall may cause flooding. Also there are other environmental factors like long continuous or rapid storms causing the amount of water to gather more and more.

Another disadvantage encountered while observing the flood prediction methods today is the dependency on computer infrastructures for performing operations. Many flood prediction models use cloud computation systems to carry out the complex computation using the inputs that are taken from observation sensors. Cloud based computation provides enough storage and power for processing the large amounts of inputs, but on the other hand, requires an available internet facility and a high bandwidth channel for transmitting data. However, in the agricultural sector at the rural level, usually there is no constant availability of high speed internet connection, thus these systems do not function to their full potential and can create a hindrance for farmers.

Also there are delays in data transmission over the internet to the cloud servers where they are being processed and results are being sent back. Delays may be caused in processing the input data as compared to predicting the flooding at the real time, as floods may be caused very suddenly and may occur with in some few moments of time which will prove detrimental to crops and live stocks.

Yet another gap that exists is that there are models that analyze parameters in certain manner without combining or showing how all of them affects the flood event. Some models will just collect the data from sensors, some may just transmit data from sensor to some observation center, while others may predict the flood using those inputs. There has been not much work done for an all-in-one model where data can be acquired, processed and analyzed, displayed to the users, and timely alerts can be issued.

Thus based on these constraints and observed drawbacks the problem can be clearly stated as; Designing and developing a complete flood prediction and alert system where:

The system is supposed to work with sensors placed at numerous places in the farm to collect data and also be able to interpret them properly. The data from these sensors are to be transmitted across large distances without need of high internet bandwidth, hence utilizing a LoRa communication protocol which will be cost effective and energy efficient for farmers at any remote locations and in agricultural zones. The analysis has to be performed locally at the gateway using the Raspberry Pi microcontroller and implemented machine learning algorithm to interpret a large set of parameters in order to predict floods. This localized prediction has to be displayed at a dashboard for the farmers (also it is important to display the past values of different parameters as well as the currently predicted value) and for distant farmers or authorities, this has to be displayed over a cloud platform. A warning or alert must be sent to the farmers if there is a prediction of flooding, and since internet access is restricted in rural areas, a GSM communication can be used to send SMS to the farmer.

"To develop an affordable and efficient IoT based flood prediction and alerting system that can monitor the multivariable sensor inputs from the farm, transmit the sensor input through LoRa, predict flood level based on machine learning on raspberry pi, analyze through a user friendly dashboard and also using a cloud based application and provide with alerts through GSM."

The proposed system will be capable of transmitting information over a long distance of 5 km with good amount of efficiency with use of Lora technology. Further with use of Raspberry Pi micro-controller the data will be processed locally at the gateway. Localized processing would reduce latency for obtaining predictions. ML would provide a robust platform for the prediction which will include multitude of parameters affecting the flood event to be predicted accurately and also make better predictions compared to just observation of water levels. Data will be visualized to be used by the farmers, and finally by using GSM network alerts can be send which would make it highly efficient for farmers at remote regions.

1.4 Research Objectives

The main goal of this research is to develop an efficient, robust and affordable system to monitor and predict flood and alert the agricultural land owners. The goals for the research are outlined as follows:

1.4.1 Design and development of the Arduino based sensor nodes to collect various environmental factors responsible for flooding such as water level, moisture in the soil and various atmospheric elements such as the temperature and humidity. The Arduino boards are selected as they are low cost, efficient and flexible to interface with the different sensors required to monitor the required parameters in the field. Collection of the real-time data in each of the fields allows the system to easily track minute changes in any of the parameters, providing an adequate input to develop a robust predictive model.

1.4.2 Establish a robust, long range wireless communication system between the sensors placed on the agricultural fields and the main processor unit, the Raspberry Pi. For this purpose LoRa (Long Range) communication modules are used to transmit data between the sensors and the main unit since it is the most suitable and economical communication method for rural and agricultural based applications to transfer data over long range and at less power. The establishment of the communication system will ensure reliable transmission of data from sensors located on wide area farms to the gateway located on the farm.

1.4.3 Design and implement data logging and cloud integration system for all the sensor nodes using the Raspberry Pi. The sensor data is received on the Raspberry Pi and can be stored locally for future use and analysis and can be simultaneously transferred to the cloud platform such as ThingSpeak for visualization. ThingSpeak is a free, easy-to-use and an open internet of things platform for sending and retrieving data from smart devices. This will allow user to have real-time access to the sensors data from the distant locations.

1.4.4 Develop a machine learning model for the classification of flood based on the data collected from the sensors on the farm and using a classifier such as a Random Forest. The training data for the Random Forest will be the various parameters being measured on the farm such as moisture, water level and atmospheric factors in the normal environment and during flood times. Random Forest is selected as the model to achieve a good prediction accuracy with the given variables.

1.4.5 Develop a web based user interface that is hosted on the Raspberry Pi for local monitoring of the farm parameters, alerts and other information. The interface would allow users on the farm to check the current state of the sensors and system status without requiring internet connectivity for monitoring the parameters and the predicted conditions on their mobile devices and laptops, without the required to login into the online cloud portal. Integration of GSM module will allow system to send an SMS on farmers mobile as the alert and for the local monitoring purpose of the sensors at the farm house without any internet connection. This will enable them to have an immediate action based on the received alerts for preventing any kind of losses in the farm and for better management of agriculture with improved efficiency.

1.4.6 Also a part of the project will include ensuring the overall system be cost-effective, scalable, and energy-efficient for the better utilization by the small farmers and agricultural producers, this will reduce the initial cost of development of the system and also the maintenance and overall system cost by efficiently utilizing resources, adding new sensor nodes easily, thus the system can be adopted in all kinds of farm with less cost. This research will ultimately help to reduce the damage and loss incurred by the agricultural fields due to the flood and also helps farmers in making strategic decisions based on the given parameters in the form of a smart farming system.

1.5 Scope of Work

The scope of work for this project covers the entire process of designing, development, implementation and testing of a flood prediction and alerting system prototype for agricultural purposes. The work mainly concentrates on building a feasible and reliable system by integrating hardware and software components to monitor environmental parameters and alert the farmers in case of imminent floods. The designed system will be a prototype that is evaluated on a laboratory setup or controlled outdoor environment. The hardware part of the project will be designing and integrating the hardware components required for the system. These hardware components will include sensors to monitor environment parameters related to flood like water levels, soil moisture, temperature and humidity. These sensors will be connected to an Arduino-based microcontroller that will be used for acquiring and preparing the data before sending it across the communication link. The designing and circuit integration will be done meticulously. Wireless communication will be a major aspect of the hardware development using LoRa communication protocol. The hardware prototype will also include designing a LoRa transmitter and receiver to facilitate communication with the central unit from the sensing node. Careful programming will be done for the LoRa modules so that the data is sent at suitable intervals.

The Raspberry Pi will also be an important hardware component. It will act as the central gateway where all the data received from the sensors will be collected and processed. The entire system can also be remotely monitored and analyzed using cloud services which can be done via the Raspberry Pi gateway. Another important aspect of the hardware would be a GSM module. This module will be interfaced with the Raspberry Pi to send SMS to the farmers as alerts about any potential floods. The local output of the system can be observed on a simple LCD display, so an LCD will be a part of the hardware as well. The software part will involve programming the Arduino and Raspberry Pi, logging the data on the Raspberry Pi, processing data, using cloud services for data monitoring (like ThingSpeak), creating a dashboard and implementing the machine learning algorithm for flood prediction.

Multiple programs will be developed for all the components, which are then integrated together. Data logging is a key aspect of the software where all the sensor values are stored locally on the Raspberry Pi. Selected data values will be sent over the internet to cloud services like ThingSpeak for data visualization. A simple dashboard will be created so that users can view the real-time sensor data and the predicted outcome with a better graphical view. This can be accessed locally via the Raspberry Pi or remotely via the cloud service. A machine learning algorithm will be implemented on the Raspberry Pi for predicting flood occurrence. A Random Forest classifier will be implemented to classify the system to be a normal or an alert situation based on the sensor data. This algorithm will be executed on the Raspberry Pi itself for efficient and faster prediction. The whole system will be tested on a controlled environment to ensure its accuracy and functionality. This will include checking all the components for proper working and response as well as verifying the machine learning prediction.

However, it should be stated clearly that developing hydrological models for large rivers is not part of the scope of work of this project. Developing such models will require vast amounts of historical data from numerous monitoring stations and complex analysis which are far beyond the scope of this prototype-based system. Also, it is not expected for this prototype to predict floods over larger time scales; it is mainly intended for imminent flood situations. The performance of the system will be measured using the data available and based on the accuracy of its prediction over the time interval that it is run.

1.6 Engineering, Environmental and Sustainability Considerations

The flood prediction system that has been proposed is designed by using cutting-edge engineering design practices, while incorporating elements of environmental consciousness and sustainability targets. The system is designed in a way that it is capable of real-time monitoring and forecasting of the impending flood situation. The system's design is fundamentally comprised of two basic parts, a transmitter unit and a receiver unit, each of which is developed with advanced engineering considerations. An Arduino Uno micro-controller acts as the fundamental part of the transmitter system and has various sensors embedded such as the ultrasonic sensor, used to monitor the water level, the DHT sensor used to monitor temperature and humidity of the environment, and a soil moisture sensor, used to monitor the saturation level of the ground, all integrated with it. Information from the above-mentioned sensors is transmitted to the receiver system by use of LoRa (long range) communication which is a type of wireless communication protocol that can transmit information over long distances with extremely low power. A Raspberry Pi is used at the receiver end to collect all the data sent from the transmitter system, store it in a data base and implement machine learning algorithms to predict flood situations and thus transmit alerts, if any.

Engineering design aspects taken into account are the accuracy and reliability of the system; the accuracy is improved by utilizing numerous sensors. Furthermore, because all information is delivered and processed in real-time, the accuracy of the prediction will be considerably higher and false alarms will be much fewer than those that could occur with any traditional prediction system. The system will be highly reliable and is being engineered to use low power devices that would transmit data without the need to worry about their functionality for a certain period of time, also the ability of it being able to handle varied conditions is taken into consideration in the engineering design phase. Machine learning algorithms that are implemented will analyze and find patterns in data to increase the prediction capabilities of the system and also improve its efficiency. The programming will mostly be done in Python, in which there are abundant capabilities that make it easy to manage data collection and also use libraries to develop machine learning algorithms, it can easily be linked with external platforms like cloud systems, or an SMS alert mechanism etc. It is designed such that components are widely available and reasonably priced; such that it can be made for mass usage especially in developing areas. One major component that makes the engineering design so efficient is the use of LoRa technology for transmitting data to the receiver system, because it has such a low power consumption and can also transmit data over a large geographical distance which can be used as an advantage to monitor a vast range of area which in itself reduces maintenance cost.

Environmental aspects considered in the designing of the flood prediction system are to protect against natural disasters, which can have dire consequences to life and property. They are among the worst natural catastrophes that have the potential to cause widespread destruction, destruction of houses, contamination of land, and the loss of human life. By means of early detection using the system, it is possible to take precautionary steps like evacuations, distribution of food, and also deployment of personnel and equipment; hence decreasing the extent of the disaster to a minimum possible level. The system's environment impact is extremely small and there are no harmful effects on the surroundings because there are no harmful emissions and also it doesn't occupy too much space, it makes proper use of electronic circuits that work on very low voltages and require less electricity. It will also play a major role in data collection and thereby in studying phenomena such as weather changes, water levels and soil saturation levels over a wide area which may prove beneficial to the long-term sustainability of the environment, thus playing a vital role in data based learning and analysis of weather patterns and thus helping us predict other nature based phenomena such as climate change.

Sustainability is an integral component in the design and application of this system. It is designed such that energy will be used efficiently and can operate on batteries and also on the alternate supply of power such as solar panels etc. The usage of LoRa ensures a massive energy reduction over other types of communication that consume large amounts of power, therefore the longevity of the batteries used is increased and thus it will require minimal maintenance. It is also designed such that it is economical for every person to buy this system and implement on a large scale such that many areas prone to floods can be protected. The design has been kept extremely simple to reduce the cost of engineering and maintenance for practical and economical implementation. It can also be expanded with more sensor networks as and when it is deemed necessary which further supports the aspect of sustainability and also ensures proper functionality of the system. It is sustainable due to its design and the use of materials and electronics which would contribute very little to landfills in the long run. The system enhances sustainability by protecting infrastructure and helping people prevent loss of lives so that they do not become a part of the socio-economically disadvantaged population group and will make the society safer by enhancing it's ability to adapt and prevent natural disasters.

The system is easily expandable to support more sensor networks in larger geographical regions, and if more advanced processing is needed for prediction then additional units that support more computation can be used such that it adds further to the capability of the prediction. Maintenance is a huge factor which increases the overall cost of system and also sustainability, so, standard components and simple architecture makes the system to be easily maintained which adds on to the sustainability factor and it is also extremely cost effective when maintenance cost is taken into consideration. This system enhances the social dimension of sustainability because it contributes to human safety and thus makes them less prone to loss of livelihood and thus more capable to take care of themselves by use of predicted warnings issued by the system to prevent devastating losses which are generally not very easily recoverable by the individual or society. It will increase awareness of natural phenomena and also enable more scientific approach towards managing environmental problems by allowing the tracking and study of environmental parameters which would normally not be feasible with existing technologies that don't focus on large scale tracking with real-time data.

Overall the system that has been proposed is an efficient combination of engineering design, environmental consciousness and sustainability principles. The system makes use of modern technology such as IoT and machine learning, has a simple modular design and real time monitoring, hence it's a practical system for flood prediction. The environmental aspects like pollution, energy conservation and data collection over a wide range are properly handled in the designing phase so as to keep the overall negative impact to the environment to a minimum and to also help in studying nature-based disasters and phenomena and predicting them at a faster and more accurate rate, also by doing so, it reduces the effect of floods and other disasters on the human life, therefore ensuring greater human safety. The system is easy to maintain and expandable such that it can be used in various regions and various capacities.

1.7 Organization of the Report

This report is structured into different chapters, in accordance with the R22 major-project guidelines of the institute. The report progresses in a systematic manner from background to the final conclusions, focusing on the respective topic of the chapter.

The report is divided into six chapters; each of which will have its own defined set of work and findings.

1.7.1 Chapter 1: Introduction

The first chapter provides the overall view of the report and its subject of the research. This chapter first deals with the problem background, importance of flood prediction in agriculture and the problems caused by traditional flood prediction techniques. It also deals with the necessity and importance of the research along with its research question. The problem statement is clearly laid out. Then research objectives of the project is stated clearly along with the scope of the project. Finally, the chapter ends by stating the structure of the report.

1.7.2 Chapter 2: Literature Survey

The second chapter involves the in-depth survey of the previously researched projects regarding flood prediction systems using the Internet of Things (IoT), including their techniques of data transmission, collection, processing, and various kinds of communication technologies that will be suitable for rural agriculture areas such as LoRa. It is also going to include the usage of embedded systems for the development of such systems like Raspberry pi, which can act as a hub for collecting and transmitting data and a brief information on different machine learning algorithm for the predictions, like Random Forest.

1.7.3 Chapter 3: Proposed system and Methodology

This chapter presents the design of the proposed system along with the required methodologies for collecting, transferring, processing and analyzing the data required for the flood prediction system. The reason for choosing the particular components in the hardware module is also discussed along with the flow of the operation using flow-charts, block diagrams, etc. The method that will be followed to implement the machine learning algorithm will be discussed thoroughly.

1.7.4 Chapter 4: System Development and implementation

This chapter deals with the implementation and development of the proposed system in detail, involving various processes like setting up of the sensors along with Arduino and connecting them to it, programming the microcontroller in its particular programming language, setting up of the LoRa module, Raspberry pi, along with its programming, and connecting of the other modules required like the GSM module to send the alerts and also the LCD screen for display. The entire implementation would include the developing of software and data storage and also the dashboard for displaying the alerts and data.

1.7.5 Chapter 5: Testing, Result and Performance analysis

The fifth chapter deals with the testing of the implemented system in different conditions to test its working. Various tests are performed to check the accuracy and reliability of the system. The accuracy of the machine learning algorithm is evaluated. The obtained results are presented in graphical format which are easily interpretable. Finally, it also explains whether the objectives of the research is achieved and how they are met along with some limitations observed.

1.7.6 Chapter 6: Conclusion and future scope

The final chapter deals with the conclusion of the study where in all the results of the system are analyzed. It discusses how the research is successful and the system developed meets the required need of the prediction system. It will also include a brief description of future scope of the project where some enhancements can be made in the system.

CHAPTER 2
LITERATURE REVIEW

2.1 Related Work

Paper 1: IoT Based Flood Monitoring System using Machine Learning Approach(Ji Da Loong et al, 2023)

This paper details an advanced IoT based flood monitoring system that integrates sensor network, long range communication and machine learning algorithms. The system uses Arduino based sensors which are configured to read environmental information like water level and rainfall. The obtained data is sent using long range LoRa communication which consumes very little power and can send data over a far distance.

One significant innovation of this study is the use of machine learning algorithms to predict floods like Random Forest, Logistic Regression. Random Forest recorded an approximate 98% accuracy, which indicates the robustness of the algorithm when dealing with multiple parameters and complex environmental data relationship. Furthermore, the system comprises of an alert mechanism including a dashboard, email alert, Telegram alert for a more user-friendly application.

There are some limitation of the system, such as it relies on a continuous reading from the sensor for predicting, if sensor reading is missed there may cause loss of accuracy of the prediction, the whole system depends on reliable communication system, sensor accuracy and maintenance is important because the performance of the system depends a lot on sensor output.

Paper 2: LoRa based IoT Sensor Node for Real time Flood Early Warning System(N.M. Yoeseph et al, 2022)

This paper reports the design of an IoT sensor node based on LoRa for flood monitoring at real time. The system uses ultrasonic sensors to acquire water level information and water-flow meters for monitoring the rate of flow of the water. LoRa is used for transmission of sensor data, as it enables data to be transferred at a very long distance.

The system managed to provide up to approximately 97% accuracy on water level measurement. It is suitable for implementation in rural and remote area.

The system suffers some drawbacks. This system focuses on measurement instead of prediction, so cannot provide warning on flood at the coming time. It has limitations to reach beyond the communication range of the gateway. The system does not consider the multiple parameter.

Here's a breakdown of the academic papers, written in a natural, human-like style, focusing on their core ideas and limitations:

Paper 3: Real-Time Flood Monitoring System Using Raspberry Pi (MFM Fairuz et al., 2020)

This paper describes a flood monitoring system built around a Raspberry Pi. They've crammed it full of sensors – water level, water flow, and raindrop sensors – to continuously keep an eye on the environment and any potential flood risks. The Raspberry Pi then processes all this information and sends it off over the internet to a server. This allows people to check the flood situation from their computers or phones through special web and mobile apps. The biggest advantage is real-time monitoring; they can see what's happening as it happens. However, this system isn't a fortune teller. It just tells you what's happening right now, not what might happen in the future, so it's more for immediate awareness than prediction. Plus, they only tested it in a lab, so it might act differently in the real world with bad weather and wonky internet connections.

Paper 4: IoT-Based Flood Monitoring and Alerting System (Mohd Fitri Mohd Yakub, 2023)

This is a straightforward, no-frills IoT flood monitor. It uses ultrasonic sensors to measure how high the water is and a NodeMCU to control things and send messages. What's cool is it has a GSM module, so it can send SMS alerts to your phone when the water gets too high, even if you don't have internet. It's simple and cheap, making it great for rural areas. The only drawback is it's really basic. It doesn't predict anything, it just reacts when the water reaches a certain level. And if there's no phone signal or the sensors get confused by wind or waves, the alerts might not get through.

Paper 5: Flood Prediction Using IoT and Artificial Neural Network with Edge Computing (E. Samitilevan et al., 2020)

This paper is all about prediction, and it's done "on the edge." They use IoT sensors to gather data, but instead of sending it all to a faraway cloud to be analyzed, they use edge devices. This means the processing happens right where the sensors are, making it much faster and good for real-time predictions. They're using Artificial Neural Networks (ANNs) to learn patterns in the data and predict floods. The big upside is speed. The downside is that edge devices aren't super powerful, so the ANNs can't be too complex. Also, network issues can still be a problem, and the sensors will need regular calibration.

Paper 6: LoRaWAN-Based Smart Sensing Unit for Flood Monitoring (M. I. Zakaria et al., 2023)

This system uses LoRaWAN, which is a type of communication that works over long distances and uses very little power. It's ideal for remote areas where internet might not be available. Sensors gather data, and it's sent out over LoRaWAN. Like some of the others, it's based on simple threshold detection – when the sensors read a certain level, it sends an alert. This is great for long-range, low-power monitoring, but again, it’s not a predictor. Signal interference can also be a problem with LoRaWAN.

Paper 7: Flood Forecasting Using Genetic Algorithm-Based Neuro-Fuzzy Network (MS Sengupta Sniroy, 2025)

This research combines genetic algorithms with neuro-fuzzy networks for flood forecasting. Basically, they're using genetic algorithms to tweak the parameters of a neuro-fuzzy system, which helps it understand the complicated relationships in water data and make pretty accurate predictions. It's a really powerful way to model complex environmental conditions. The catch? It needs a lot of computing power and takes a long time to train, which isn’t ideal if you’re trying to build a cheap IoT device for remote areas.

Paper 8: Machine Learning-Based Flood Severity Prediction Using IoT (Devendran P. Et al., 2023)

This system uses machine learning, specifically wavelet transforms and adaptive neural networks, to figure out how bad a flood is – not just if there’s one. It collects data from IoT sensors and then uses it to help people make decisions. Being able to classify flood severity is a big plus. The downside is that the data needs to be clean; if it's messy, the accuracy suffers. Also, the system relies on some pre-programmed rules from experts, which adds to its complexity.

Paper 9: IoT-Based Flood Prediction Using Dam Monitoring (Shrishail S. Patil et al., 2023)

This one is specific: it's all about monitoring dams using IoT sensors and a decision-making model called TOPSIS to analyze multiple parameters and issue flood warnings. It seems good for dam safety but depends on SMS alerts and will need frequent sensor maintenance. Its accuracy might take a hit during intense rain.

Paper 10: IoT + LSTM-Based Flash Flood Detection System (Khaled M. Abdelmagid et al., 2025)

This is a more advanced prediction system using LSTM (Long Short-Term Memory) networks, which are great at recognizing patterns in time-series data. It can predict how severe a flood will be and classify it at different levels. The main advantage is improved accuracy due to its ability to learn from past data. The downsides are that it needs a lot of data and computing power, and any break in the continuous data stream will mess with its predictions.

2.2 Comparison of Methods / Approaches

Several methods and approaches have been implemented or developed over the years to enable floods to be monitored and predicted. All these techniques have advantages and disadvantage to some extent. These techniques ranged from manual observation systems to advanced technology-driven solutions that incorporate the use of IoT and machine learning. A comparative study of these approaches can help to identify the strength and weakness, and to indicate the need for an enhanced and integrated system for agricultural flood prediction.

Traditional method for monitoring flood involves monitoring through manual observations, measuring water level using gauge, observing surroundings, etc. This method is low cost, and does not require any special infrastructure but these are manual processes, thus information are collected at specific time and cannot be relied upon, also lack in automatic notification features.

Automated monitoring systems involve usage of electronic devices. These devices use sensors to monitor the parameters like water level and rainfall and automatically sends an alert when it reaches a threshold. Such systems still lack in prediction and hence it cannot forecast about the potential danger. These systems require additional support systems like communication networks and power source. However it overcomes the problem of human monitoring and thus timely alerts can be sent. However most of these automated systems make use of threshold based calculations and do not effectively capture the complexity involved in flood scenario with interaction of different factors, hence there might be issue like false alert or failure to send alert under certain circumstances.

The Internet of Things (IoT) system for flood detection involves a network of interlinked sensors and devices that collect and transmit real-time data, allowing farmers to monitor environment remotely through cloud. With such efficient systems for gathering instantaneous and comprehensive information on environmental status, better monitoring and informed decision can be made. These systems typically use communication technology such as Wi-Fi, GSM, or LoRa for transmitting data from the sensor nodes. However most of these systems depend on internet availability for transferring data and storing the data in cloud for its computation and processing. Therefore the system will have performance issues in areas with poor connectivity such as rural or agricultural region where there is lack of internet connectivity.

Machine learning approach involve training of algorithm using historical and real-time data of different parameters, the algorithm learns from these parameters and make predictions. These methods help to develop sophisticated systems which are more accurate than threshold based systems. Various techniques of machine learning that has been used are SVM (Support Vector Machines), ANN (Artificial Neural Network), Long Short-Term Memory (LSTM) and Random Forest Classifier. However large training datasets are required to train such systems, which are not always available in rural and agricultural region, further these algorithms consume more computational power and time, hence they often require usage of cloud, which may again be problematic in remote areas.

The system proposed herein, integrates both the capabilities of IoT and machine learning to mitigate some of the challenges associated with the individually based systems. The system utilizes IoT sensors to acquire data from multiple parameters, while the predicted outputs were obtained using machine learning (Random Forest Classifier) to accurately predict flood occurrence. With the use of IoT based sensors with data collected in real-time on one hand, accurate flood predictions were made using data trained machine learning models, on the other hand. To overcome the drawbacks of these methods that often use GSM, Wi-Fi or Internet connectivity, which is unreliable in remote area; the LoRa (Long Range) technology is utilized, which provides a long distance communication with lower power, also it has capacity for robust andreliable data transmission in remote areas and has an integrated alerting system using GSM. Raspberry Pi is utilized to carry out the task on edge, to avoid latency, and to utilize the computed parameters from different sensors without being completely dependent on the cloud, therefore minimizing cost.

Table 2.1 – Comparison of Methods / Approaches

Method	Technology Used	Advantages	Limitations
Traditional	Manual / Sensors	Simple, low cost	No prediction, delayed alerts
IoT-Based	Sensors + Cloud	Real-time monitoring	Internet dependency
ML-Based	Data + Algorithms	High prediction accuracy	Requires large dataset
Proposed System	IoT + ML + LoRa	Predictive, low cost, real-time	Prototype stage

2.3 Critical Analysis of Literature

Analysis of the existing research works and scientific articles concerning flood monitoring and prediction demonstrates that despite enormous technological advancements; there are several issues that the existing flood prediction systems possess. These issues are the primary reasons why the existing systems of flood prediction is neither very effective, nor easily deployable and accessible for agricultural or rural regions. Pinpointing these issues helps to formulate a system which is capable of mitigating these shortcomings.
One of the major limitations that the existing systems possess is the lack of real-time observation and prediction facilities. Current systems use manual observations and simpler measurements of water levels. The nature of traditional flood monitoring is entirely reliant on the human efforts which leads to latency in data measurement, processing and warning which consequently means the warning is only generated after critically reached flood situation and can minimize damage from a flood that has already begun but is not at all effective in terms of prediction and mitigating agriculture associated issues. This makes the system inefficient in agricultural context and not efficient enough to prevent the crops from being destroyed.

The introduction of Internet of things (IoT) technologies have been introduced that has solved many issues in relation with real-time observation and observation from a distant location. These systems integrate a series of sensors in the form of a network which collects real-time environmental parameters such as water level, moisture level, rainfall intensity, temperature etc and transmit to the cloud. The end-user can monitor it through an internet platform providing enhanced situational awareness and immediate response. However, there are still several drawbacks in the IoT-based systems such as dependency on internet for transmitting the data and to process the data at the cloud, hence these are not applicable in the rural/agricultural regions which lacks basic internet connectivity or are completely cut off. Furthermore these systems will pose increased operational costs due to reliance on cloud infrastructure, and will result in the time lag during processing and transmission of data and alerts.

Another main advancement in the flood prediction systems that is prevalent these days is based on machine learning algorithm. Machine learning algorithms are implemented to identify patterns in large scale time-series data, historical and real-time data sets and hence forecast future behavior based on current predictions. Machine learning algorithms such as Support Vector Machines (SVM), Artificial Neural Networks (ANN), Long Short-Term Memory (LSTM) etc can forecast floods with very high accuracy. The problem with such an algorithm is the vast amount of training data that is required and this might be a serious problem in agricultural background as complete relevant data may not always be available and it can also result in over fitting the model to certain particular patterns which may not occur in other circumstances. Moreover, advanced machine learning algorithms requires large processing capacity which would require external computing capabilities to run which cannot be accommodated within the low-cost system and within embedded device.

Another major gap identified in the existing systems is based on simulated systems that are designed and operated primarily in a controlled and a designed environment without actual physical operation. Various papers are only providing simulation of system which shows excellent performance in that condition but not in actual application where there might be issues such as noise, wrong sensor value, communication failure and external factors.

Additionally, majority of existing systems address either the sensor network alone or the alert and warning system. Certain systems only focuses on data transmission using available infrastructure. This is not a complete flood monitoring system. All the component required to effectively warn and monitor floods have not been integrated into a system. Therefore, a complete system is a system that should consist of the entire path from the sensors to alert.

Most existing systems lack the ability to scale according to the needs and requirements of any given terrain and the high cost associated with them do not make them accessible to farmers from small and marginal agricultural lands who are worst hit by the calamities. It is imperative that the system has the capability of scalability for different types of environment along with affordability and simplicity.

The research gap, therefore, is in the system that needs to be created by integrating the above mentioned components as a one holistic system, which should possess both the local data storage, analysis and prediction with low cost embedded systems without heavy reliance on external infrastructure such as internet connectivity and powerful computing sources that could provide high degree of reliability and efficiency even in the rural context.

The proposed research is going to try to develop an intelligent IoT based flood prediction and alerting system with machine learning to effectively address the problems mentioned above with enhanced accuracy and efficiency through long range communication, edge computing and integration of entire system with lower cost of installation and operation for rural use.

Hence in a Nutshell it can be stated that lot has been researched in the field of flood prediction but there is a need to create system that integrates few advanced technologies that could address the problems such as, accuracy, cost of installation and maintenance, effectiveness, reliability and efficiency while ensuring adequate scalability, accessibility, and applicability for a better and safer society.

2.4 Research Gaps Identified

Based on a deep and critical analysis of several researches and existing systems within the flood monitoring and prediction domain, many crucial research gaps were identified. These gaps indicate limitations within current research endeavors and strongly recommend the adoption of a more practical, efficient, and integrated solution particularly within agricultural applications in rural environments.

The foremost among these gaps is the lack of an affordable and scalable flood prediction system for rural farming areas. In the mentioned researches, the systems have been developed with very expensive components like high end sensors, advanced communication network and cloud based computing which would be difficult for small and marginal farmers to purchase and maintain. In an agricultural country like India, farming is a common livelihood, and not all farmers can afford such high tech systems. This shows that a cheap and scalable system for large rural farms is missing in the market today.

Another critical research gap is that most of the traditional systems only depend on one single parameter like the rainfall, or water level to predict flood conditions. For flood prediction, soil conditions, water table, and temperatures are as important as the amount of rain falling and water flow. These all interconnected factors, are many of the time ignored, reducing the prediction accuracy and reliability. Thus, there is need to implement a system that can monitor and analyze multiple parameters simultaneously.

A further gap within existing research systems is the insufficient use of machine learning implemented at the edge of networks. Majority of present systems use cloud based computing facilities to calculate prediction. Although they give high prediction accuracy, but are dependent on the Internet connectivity which is not feasible for rural areas. Also, transmission of data and retrieval of prediction leads to delays. With the development of the edge computing, more machine learning models need to be implemented at the edge itself. For example, using systems like Raspberry Pi. With local processing, a lot of delay can be avoided and also it is more efficient when there is intermittent or no internet connectivity.

One of the major issues identified in flood monitoring systems is that most systems are only able to notify through internet based services. In many rural agricultural regions, farmers may not possess a smartphone or access to the internet to get a notification. Therefore, the best solution for such communities is an SMS based warning system. While some systems can warn farmers of flood prediction, there are very few that can send alerts via SMS.

In addition to these gaps, there is also the lack of end-to-end integrated systems. A few systems exist which can perform tasks like data sensing and transmission, while others focus more on prediction only. Most of these existing systems can perform certain set of operations but they are not integrated to form one complete solution that performs sensing to warning.

It has been observed that most researches have conducted experiments in the laboratory or used simulations only. In the real agricultural environments, the factors which influence the system like variations in climate, sensor failures and communication are different. So, it is very important to test the developed systems under the actual agricultural conditions.

The presented solution aims to bridge all these gaps by providing an end to end smart flood prediction system which is inexpensive and robust, which could provide warnings to small and marginal farmers even in remote locations and also provide the prediction with the minimum latency. The proposed system has been developed using IoT technology, LoRa for data transmission, Raspberry Pi for edge computation using Random Forest machine learning model and a GSM module for alerting.

CHAPTER 3

APPROACH / DESIGN METHODOLOGY

3.1 Overview of Phase–I Work and Outcomes

Phase-I of the project is to lay a concrete foundation of the development of the proposed flood prediction & alerting system. It was important in this stage to analyze the problem domain & identify shortcomings of existing solutions & formulate an approach that effectively combats problems of flood monitoring in agricultural scenario. Phase-I involved many sub-activities for this purpose, which is going to be very useful for system design & development stages in latter phase of project.

First of all, in phase-I, intensive problem domain study was performed. Flood problem for agricultural land was studied from both the environmental & technical point of views. Effect of flood to crop yield, soil quality and farmer's life was determined to emphasize the importance of the problem. Factors for flooding such as rainfall quantity, poor drainage, water table and soil saturation, rising level of adjacent water bodies, were studied.

Another significant step done was thorough literature review. Many articles related to flood prediction & monitoring were studied. All of them were classified in three main categories, traditional method, IOT based system and machine learning based system. Conventional method for prediction was simple and cheap, but it did not had any automated or prediction capability. IOT based system used many sensors for continuous real time monitoring, but the disadvantage of system were need of stable internet connection and use of cloud infrastructure for computation and data storage. Machine learning based system was very efficient for prediction, but it required large amounts of data to predict and requires more computational power.

It was found that present systems have three shortcomings: no multi-parameter analysis is used, system relies on cloud computation and no integrated alerts system is used for informing people about the threat.

With an intent to develop a comprehensive flood detection and alerting system for agriculture environment, need of development of integrated system using both IOT and Machine Learning technologies was established.

Feasibility studies of using low cost and easily available hardware were conducted during the project phase-I. For this, Arduino UNO board and Raspberry Pi were selected. Both are affordable, easy to use and available in market. For wireless data transmission across long distance and low power, LoRa (Long Range communication) was decided.

To analyze feasibility for practical implementation, a small experiment for testing the basic functions of sensors like water level and soil moisture sensors was designed and implemented by interfacing with Arduino board.

Besides testing sensor performance, for the preliminary work data of many parameters were collected for understanding relation between various parameters like soil moisture and water table. This data was collected and stored for developing models in later stages of the project.

Last of all the system constraints and implementation issues were identified and defined. These were sensor range, accuracy, communication issues, data usage and cost issues.

Based on the feedback received from the Phase-I evaluation of the project, a few essential improvements and refinements were made to better enhance the effectiveness and feasibility of the flood prediction system. The Phase-I evaluation of the project mostly included the understanding of the problem, survey of existing systems and a basic conceptual model. The project evaluation session has, however provided a useful insight into the loopholes of the system design at Phase-I level and, therefore, certain necessary changes have been incorporated in the system design and objectives to achieve the desired level of accuracy and feasibility of the system in agricultural surroundings.

At the Phase-I level the system design was quite rudimentary in terms of monitoring the environment conditions like water level and rainfall level, the water level and rainfall was sensed and based on predefined threshold value, the warning was issued for flood detection. The threshold level could be defined as the limit of water level and rainfall, which can be taken as an indicator of flood occurrence. Although this was easy to implement, it suffers from limitations of prediction accuracy.

At the evaluation phase it was discovered that the water level and rainfall conditions is not a sufficient indicator of flood conditions as there are number of other parameters involved in flood conditions. For example if the ground level is already soaked up with water, it will be quite prone for a flood at even minimal rain. Conversely, if the ground is dry it would take much higher amount of rain to cause flood. Hence use of a single or two parameters for flood detection has certain limitation which will result in the erroneous prediction of floods. Thus it was found that to have the efficient and more accurate prediction a greater variety of environmental parameters needs to be introduced in to the system. Accordingly, a higher variety of sensors were used for data acquisition, for example, water level sensor, rainfall sensor, soil moisture sensor, temperature sensor and humidity sensor. Out of all the parameters soil moisture sensor plays a critical role in determining the amount of water that could be held by the soil before it causes the run off and then flood. Similarly the parameters like temperature and humidity also plays a crucial role in analyzing the soil moisture levels by affect the rate of evaporation. Thus the amount of data from various sensor can lead to a much efficient flood prediction system.

At the Phase-I level traditional communication modes for data transfer was to be the part of the system like WiFi. The WiFi protocol suffers from many drawbacks which is unsuitable for application in agricultural settings like relatively lower transmission range and higher power consumption. Moreover there is a problem with the reliability of connectivity in agricultural fields. For the above said reasons the system designers decided to adopt a more energy efficient and long range communication mode called LoRa (Long Range Communication) protocol. This protocol is designed to allow data to be transmitted for longer distance with minimal power.

As compared to the traditional system of warning based on threshold values, where if certain parameter exceeded certain defined limits the warning was issued, the present system incorporates more intelligent approach using a machine learning based approach to overcome the issue. Machine learning helps the system to study and understand the relation between various parameters and thereby make a much more efficient prediction of potential floods based on machine learning techniques like Random Forest algorithm. The Random Forest is a classification and regression algorithm that operates by constructing a multitude of decision trees at training time.

The proposed objectives were improved so that the system can incorporate prediction, which not only includes detection of flood but also prediction based on analyzing the pattern between the environmental parameters over a period of time. Other than prediction it was also decided to analyze the parameters that contribute to the flood and also to analyze the individual contribution of every parameter. Furthermore a real-time alert was introduced so that the detection of flood can be easily and effectively transmitted to the users of the system through reliable methods like GSM based SMS.

3.2 Refinement of Problem and Objectives Based on Feedback

Based on the feedback received during phase-I evaluation some modifications and improvements have been made to the proposed flood prediction system to increase the functionality and practicability of the system. Originally the proposed system was too simple and just consisted of monitoring the rainfall and water level sensor data using simple threshold based logic. In this system some threshold values are fixed and whenever the values from the sensor become greater than the defined threshold the system generates a notification. However, this type of approach cannot determine the accurate flood condition because the occurrence of flood depends on multiple environmental factors that are interconnected.

One of the recommendations in the phase-I evaluation was that there should be inclusion of some more parameters in the system in addition to water level and rainfall data. The system must take some more parameters like soil moisture, temperature, humidity etc. For predicting the flood. If the soil moisture is already saturated then with even less amount of rain the system may experience flood condition whereas heavy rainfall may not lead to flood if the soil has higher absorbing capacity. Thus to enhance the accuracy of the flood prediction system some additional sensors for soil moisture, temperature and humidity are incorporated in the proposed system.

Another important modification made to the system based on the feedback received was regarding the communication protocol. Earlier for the communication of sensor data a traditional network protocol was decided to be used such as WiFi. However, the system based on WiFi will have low coverage range, high power consumption and it is dependent on the internet for transmitting data. In agriculture scenario especially in rural areas where internet connection is not easily available it will not be suitable. Therefore, in the present modification the system uses LoRa for the transmission of data which is Long Range. It provides coverage up to 10 KMs even at a single chip power level of below 20 mW without internet connectivity.

A significant modification that was made to the system during this stage was that instead of just detecting the flood based on threshold the system will use the machine learning to predict flood. The system now uses Random Forest algorithm to determine flood conditions by predicting it from sensor data values. Machine learning is a method by which we can increase the accuracy of the prediction using the relations between multiple parameters and hence it can predict the conditions more efficiently. The Random Forest algorithm is used for its better prediction capabilities.

There was also expansion made to the objectives of the proposed project. Originally the objective was only monitoring and giving an alarm but according to the recommendations of phase-I, there were a number of objectives added for increasing the usability of the system. The system will continuously collect and monitor parameters and will provide prediction about potential flood in real time and this may be done dynamically as the system is based on the predictions from the machine learning.

One of the other objectives which have been added is that system will be giving the prediction on the individual parameter's basis. It will analyze the individual parameter's role in creating the flood condition.

Third more objective has been added for the proper alert mechanism to make the system more user-friendly. The system now gives the alarm in the form of SMS alerts via GSM module and by updating the same to the dashboard.

Hence it is concluded from the modification made at this stage that the flood prediction system has become a more intelligent and efficient one which predicts the future conditions more efficiently and hence it would be very useful for agricultural applications.

3.3 Selection of Approach / Methodology

The selection of an appropriate methodology is a critical part of any engineering design, which directly impacts on the system performance, reliability and feasibility.

For this project of flood prediction and warning system in the agricultural sector, selection of the methodology required the assessment on some important factors such as the cost, reliability, power consumption, speed and suitability for the rural applications. Based on the investigation of other approaches and its disadvantages, a hybrid methodology of Internet of Things (IoT) and machine learning is proposed for the system.

For the purpose of flood prediction and warning in agricultural environment, a conventional flood monitoring system based on manual method and simple threshold method is not adequate because it does not possess the predicting capability. While the IoT system could be utilized to collect and monitor the data in real-time but often lacks the ability of prediction. Machine learning could also provide the high accuracy and efficiency in prediction but needs a sufficient volume of data. Therefore, to overcome limitations posed by the two standalone methods, a hybrid approach which combined the IoT system with the machine learning system has been chosen for this project.

The IoT component plays an essential part in data collection from different sources of the environment, especially for an agricultural region where vast amount of environmental data is needed. With the aid of sensors that can measure parameters relating to flood events, such as water level, soil moisture, temperature, humidity, and rainfall, the data can be obtained in real-time, and forwarded to the microcontroller at regular intervals and eventually to the central processing unit. It is a vital component in creating the flood prediction system as it is continuously gathering and relaying information, and it has been utilized with a microcontroller.

The data thus collected has been processed with machine learning to forecast possible future flooding scenarios. The collected data are transmitted wirelessly to the local processing unit in order to achieve long range transmission with minimal power consumption through a wireless module called LoRa (Long Range) technology. Wi-Fi and cellular technologies often require a readily available network, which can be expensive and may be unavailable in rural and remote agricultural regions. The selection of LoRa provides cost-effective communication for a vast area of application. This is one of the most significant advantage of this system especially when compared to existing solutions.

In the machine learning component, sophisticated classification algorithm was utilized for making a decision. Among various machine learning techniques, Random Forest Classifier was chosen for the project due to its higher accuracy and robustness as a classification tool. Random Forest combines multiple decision trees during training phase and combines the output from the decision trees. The ensemble technique utilized by Random Forest helps reducing the over-fitting problem, enhances the accuracy, robustness and generalization performance of the model. Non-linear relationships can be discovered, which are crucial in forecasting natural phenomena like flood as they are usually influenced by non-linear interactions between variables.

Also, machine learning classifier like Random Forest allows using different input types, as it is compatible to work with numerical and categorical features. Feature extraction and important variables for determining flood occurrences are among the advantage that can be achieved using Random Forest algorithm. Also it demands lesser parameter tuning and is suitable to be implemented on embedded devices like Raspberry Pi.

The learning approach selected uses local processing unit (Edge Computing) instead of cloud-based learning which will cause higher response time and dependency on network availability. Learning process is performed locally at Raspberry Pi, a single-board computer, thus resulting in lower power consumption. An accurate warning system with lower response time will contribute directly to minimize losses. This methodology can lead to an intelligent monitoring system, which not only detects flood based on various parameters but also predicts its occurrence and takes preventative actions.

To make the system efficient in alerting the users in event of prediction of flood, SMS alerts are sent using a GSM module. In the absence of network connection or with unreliable network, farmers can still be alerted without the dependency on mobile network. An example of alert being generated from a prediction is "Warning! Flood prediction. Water level expected to rise by 0.5 meters within 2 hours". This notification has been relayed using SMS service, making the system efficient in notifying farmers.

The use of the cost-effective components like Arduino, Raspberry Pi, and the low cost of LoRa module will lead to an affordable and scalable flood warning and prediction system that is suitable for small and marginal farmers in the agricultural domain. The scalability can be further enhanced by using additional sensor nodes to cover larger areas of farmland.

From the above discussions, it can be inferred that the selected methodology has been appropriately devised to develop a reliable, robust, cost-effective, power-efficient and a highly scalable flood warning and prediction system by utilizing the benefits of IoT and machine learning with efficient wireless communication and edge processing.

3.4 Technical Approach and Key Elements

The proposed flood prediction system relies on a layered approach where different parts perform different functions for the system to work correctly. This organization not only helps the system remain structured but also make it easier to scale up and implement. It is divided into four layers; sensor layer, communication layer, processing layer and alerting layer. These are crucial components in enabling the system to monitor the surroundings, evaluate the gathered data, forecast flood risks and promptly inform users of any potential danger.

Sensing Layer:

The sensing layer is the fundamental aspect of the system as it is used for gathering real time environmental data. This layer comprises of various sensors connected to the Arduino. These sensors are located in agricultural fields, monitoring continuously different aspects of the environment that can be either directly or indirectly related to flooding.

The primary parameters to be measured by these sensors are water level, rainfall intensity, soil moisture and temperature. A water level sensor in proximity to rivers and streams will alert the farmer about the potential for a flood, indicating the rate at which the water is rising. Rainfall sensors will gather data on precipitation levels to determine the rate at which the water is increasing. Soil moisture sensors can inform about the volume of water content within the agricultural land. An optional temperature sensor can also be added, the values of which may correlate to changes in the environment leading to floods.

These sensors are connected to the Arduino uno. The Arduino is a relatively simple and versatile board used for data collection. In this system, Arduino is programmed to continuously read data from these sensors. This collected data is then to be read by another component within the system and transferred elsewhere for further processing. The system has been designed for low power usage in mind to accommodate deployment in agricultural settings far from electricity power sources.

Communication Layer:

This layer will be responsible for relaying the data collected from the sensors to the central processing unit (CPU). The data will be communicated using the Long Range Radio (LoRa) technology, specifically suited for the rural and agricultural environment as it requires very low power and can transmit data over a relatively large distance.

In this system, the Arduino board is attached to the LoRa transmit module to pass data on to a gateway. The gateway is in this system will be a Raspberry Pi fitted with a LoRa receive module. This enables the two to communicate wirelessly for over many kilometers thus removing the need for cables. An advantage to using LoRa technology is that it doesn't require an internet connection. This will be advantageous as mobile networks and infrastructure for data transfer over the internet may be scarce in agricultural regions. Data sent from the sensor node will eventually arrive at the Raspberry Pi. Once at the Raspberry Pi, the data will be stored locally typically in a .csv file. This file will be used for input to the machine learning model.

Processing Layer:

The processing layer is responsible for analyzing the data obtained and predicting any potential floods. This part of the system will be done using the Raspberry Pi which acts as the CPU. The data will be inputted to a machine learning algorithm that will be trained and deployed to distinguish between safe conditions and a high flood risk.

This system will employ the Random Forest algorithm due to its robust nature and high predictive accuracy on multi dimensional data. It achieves its performance by creating multiple decision trees in a way that combines several predictions from decision trees. Data will be gathered from the .csv file for use in training the model as well as its deployment phase. The trained model will be utilized to predict if there is a high flood risk with the real-time data from the sensors. Advantageously, the use of a Raspberry Pi on site utilizes the concept of edge computing. This minimizes reliance on cloud computation and significantly speeds up processing time, as speed is essential in flood prediction systems.

Alerting Layer:

This layer will communicate any prediction to users, including farmers and local stakeholders. This layer involves visual aids and actual alerts. A web based dashboard will show real-time data. The display will allow the user to monitor weather data and see the predictive capabilities for any flood threat. This can be accessed by the user anywhere they have internet access or through the Raspberry Pi itself if on site, enabling immediate action. SMS messages will also be sent to a user's mobile phone using a GSM module. This provides a guaranteed means of communication to the user, especially where data coverage is not a luxury. This alerting system ensures that all users are well-informed so that they can initiate precautions against impending flood conditions such as protecting crops and livestock, draining lands that are likely to be submerged, and moving any high-value assets that may be endangered.

Overall Integration:

All the layers are integrated into a complete flood prediction system. The sensing layer gathers data from the field. The communication layer transfers this data. The processing layer predicts the threat of floods. The alerting layer informs users about any threats predicted. Together, these four layers present a complete package that guarantees continuous monitoring, precise prediction and timely alerts to reduce the risks to the user.

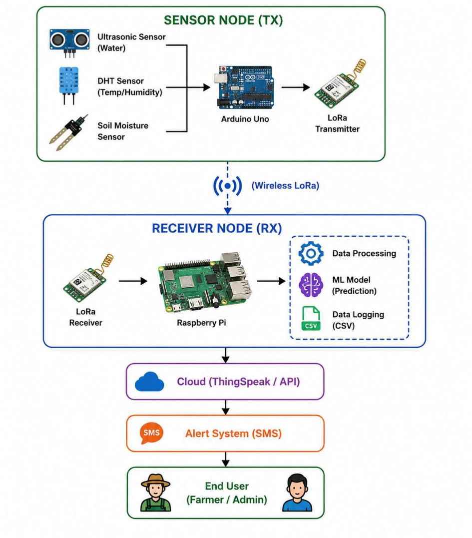


Fig 3.1 System Workflow Representation

3.5 Supporting Analysis / Design Considerations

A number of key considerations were applied to ensure the successful and practical design of the proposed flood prediction and alert system. In consideration of the intended agricultural environments, several technical and practical considerations have been addressed within the design of the system to ensure it is resilient and effective, particularly in rural and remote regions where access may be limited and technology implementation has unique challenges.

Power consumption was considered one of the highest priorities during design. With potential access limitations to stable and consistent power in remote agricultural locations, it was paramount that the system operated on the lowest power possible, thereby extending its lifetime on its battery source or by drawing minimal power from renewable energy sources such as solar panels. The utilization of low-power technologies such as LoRa radio modules has drastically reduced the system's power consumption when compared to alternative wireless protocols such as Wi-Fi which are more power hungry and typically require a continuous connection to an internet source. LoRa was specifically selected as it is designed to send small amounts of data over long distances on low power, the ideal solution for sensor nodes in remote locations.

Accuracy and reliability of sensors is one of the critical factors governing the effectiveness of the system, as it directly feeds into the predictive accuracy of the machine learning algorithm. Therefore, a strong emphasis was placed on selecting high-quality sensors which would be accurate and stable over time. Calibration of these sensors was another crucial element that needed to be addressed, as well as precise positioning to collect reliable data for analysis. The quality of sensor input is integral to the machine learning component of the system as flawed input data directly correlates to flawed output and predictions.

The stability and reliability of the communication protocol are fundamental to a distributed sensor network. Data transfer is essential from sensors to the central processing unit and loss of data due to poor communication leads to diminished performance and reliability of the system. For this reason the LoRa (Long Range communication) technology was chosen, it offers superior range over traditional wireless communication, and also provides reliability for transmitting data over many miles, with trees and buildings potentially impeding radio waves, making it ideal for agricultural land. Additionally, it is not dependent on an internet connection which adds to its reliability in areas with limited network coverage, and is extremely low-power, helping it conserve its power supply.

Cost effectiveness is an integral part of implementing any new technology in the farming industry, as farmers are not always financially positioned to implement costly systems. Therefore, this system was designed to be relatively inexpensive without compromising on features and efficiency, using easily sourced and affordable hardware such as Arduino Uno, Raspberry Pi and the machine learning algorithm. This greatly reduces the overall cost, while also ensuring that individual components are readily available for replacement should they fail.

Internet independence was considered when designing the system as there are still many areas which are difficult to access by reliable and continuous internet coverage. As most current IoT systems rely on the cloud to store and process data, an internet connection is necessary for the system to function. In an agricultural context in a remote rural area this poses a large issue and the system was therefore designed using edge computing, whereby the processing is performed locally. In this setup, data is collected locally on a device like the Raspberry Pi, which can then analyze the information and create predictions locally without the use of the internet, only sending out SMS alerts when necessary using the GSM communication system, further increasing the system's overall independence and effectiveness.

Scalability was another key aspect that was taken into account during the design process. As an expanding industry, it was considered that the system may be used to monitor larger areas or perhaps different parameters and therefore the system must be adaptable. Through the use of modular design it is simple to add more sensors, either to increase the area of coverage for an existing prediction, or to include different parameters that should be monitored and analyzed.

Robustness and durability of the system was considered paramount for an outdoor sensor system. The systems are designed to be weather-proof, and must be able to withstand temperatures, moisture, dust, and any other factors which may affect the performance of the internal components. In terms of functionality, the system must be robust and able to handle potential failure of a sensor or a communications issue and still operate.

Efficiency of the data processing and prediction step of the algorithm was taken into consideration to be able to run off less powerful hardware such as a Raspberry Pi. The Random Forest algorithm was chosen for the prediction process as it requires far less computational power than many other algorithms such as neural networks, allowing for it to be implemented successfully without requiring excessive computational resources or significantly depleting battery power.

Finally, the usability and accessibility were taken into consideration. The system was designed to be straightforward to implement, and the outputs, either through the display of data on the Raspberry Pi or through the dispatching of SMS alerts, would be easily accessible to a farmer who is not necessarily technically gifted. This aspect will increase adoption and effectiveness.

3.6 Tools / Techniques Used

The developed flood prediction and alerting system has to make use of many different tools and techniques in different fields; from hardware realization to software, from communication to data processing and machine learning. Each of these tools and techniques was selected with great care, considering its functionalities, economic feasibility, usage complexity and appropriateness in an agricultural environment. It is these different tools and techniques that constitute the heart of the system and make possible: sensing, transmission, processing, prediction, display and alerting.

To simplify and clarify the different tool and their uses, here follows a table with the different techniques implemented in the project:

Table 3.1 – Tools and Technologies Used

Tool / Technology	Purpose
Arduino IDE	Programming Arduino sensors
Python	Data processing and ML model
Raspberry Pi	Gateway processing unit
ThingSpeak	Cloud data visualization
LoRa Module	Long-range communication
GSM Module	SMS alert system
Scikit-learn	Machine learning implementation
CSV Files	Data storage and dataset

Detailed Explanation of Tools and Techniques:

Arduino IDE

One of the most important parts of the project is the Arduino IDE because it is the interface in which the code for programming the Arduino microcontroller is written, and it is the Arduino microcontroller which gathers information from all of the sensors. Simply speaking, it is where the code which defines how the Arduino will function is written. The IDE allows code to be written and then sent directly to the Arduino and is a very convenient piece of software for working with sensor applications. One of the key reasons that the Arduino IDE was chosen was because it is very simple to use, even for someone who has had no previous experience coding. The IDE consists of a basic text editor in which code is written in C/C++ (otherwise known as the Arduino language), and built in libraries for communicating with the hardware (such as sensors). This lowers the difficulty involved in coding. In the context of this project, the Arduino IDE is used to communicate with different environmental sensors such as water level sensors, soil moisture sensors, temperature sensors and humidity sensors. These sensors transmit different pieces of raw data that must be read by the Arduino. The code in the Arduino IDE outlines how these different sensors are read and how the information given by the sensors is used. For example, the reading given by a water level sensor or a soil moisture sensor will likely be an analog reading that must be converted into a digital reading to be usable. Another benefit of the Arduino IDE is serial communication. This is an important feature for the development and testing process as the Arduino's serial monitor can be used to display any data sent by the sensor, which may indicate how it is functioning. In other words, problems may be noticed by looking at the serial monitor. There is also a vast number of libraries for the Arduino IDE which allows for simplified use of various sensors; rather than writing code for the sensors from scratch. There are many libraries available for nearly every sensor and this makes development of systems such as this much easier. Finally, the Arduino IDE is available for a wide variety of operating systems such as Windows, Linux and MAC, and is freely available and open-source. In short, the Arduino IDE is vital to the development of the sensors for this project, as it gives access to powerful yet relatively easy to use functionality which streamlines the sensing process.

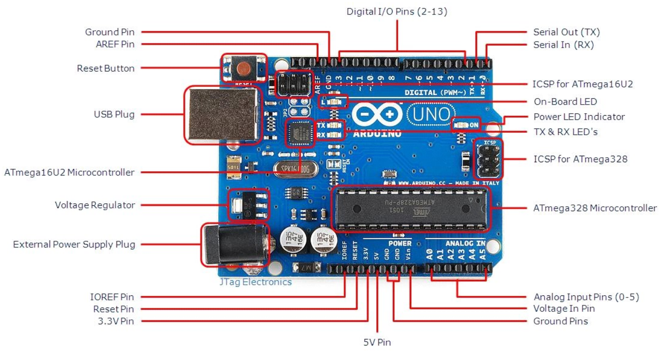


Fig 3.2 Arduino

Python

This project primarily uses Python for data processing and machine learning implementation. It is the primary component for transforming raw sensor data into informative and predictive information. One of the main reasons Python was selected is its straightforward nature and readability, making it accessible for developers of all skill levels.

On the Raspberry Pi, Python is used for numerous tasks, including the reception of data from the LoRa module. After data has been collected by the LoRa module, it is processed using Python scripts and stored in a structured format such as a CSV file to make it ready for subsequent analysis. Python is able to manage the amount of data being passed efficiently.

Data preprocessing is another significant application of Python in this project, whereby raw data is cleaned and formatted before it is fed to the machine learning model. The Python libraries for data manipulation include NumPy and Pandas, which simplify the management of data by allowing the sorting, filtering, and manipulation of data elements.

Machine learning model implementation in the system is another area that benefits from the use of Python, enabling the seamless integration of algorithms into the project. It analyzes sensor data and predicts the probability of a flood occurring. Python scripts run on the Raspberry Pi to manage incoming data and provide real-time updates of predictions.

The use of Python provides robust support from its extensive libraries, such as Scikit-learn, Pandas, and Matplotlib, that provide tools for data analysis, machine learning, and data visualization, thereby saving the user considerable time writing complex code.

In the context of the alerting system, Python scripts initiate communication with the GSM module for sending SMS messages in the event of a predicted flood. Users are alerted instantly, allowing them to take preventive measures if necessary.

In terms of portability, Python is capable of running on different systems including the Raspberry Pi, thereby making it suitable for edge computing environments and embedded systems. In addition, its ability to perform multiple tasks concurrently makes it an effective tool for system component integration.

In conclusion, Python is a central feature of the system, forming the foundation for data processing, machine learning, and communication, as well as being versatile and effective in handling complex computations thanks to its readily available libraries, making it suitable for implementing an intelligent flood prediction system.

Raspberry Pi

The Raspberry Pi is used as the central processing unit, essentially the 'brain', of the entire system, which is responsible for processing the sensor data received from the sensor nodes, executing the machine learning algorithms, and initiating any alerts; consequently, it is integral in bringing all of the system components together.

One of the primary benefits of using a Raspberry Pi is its relatively inexpensive price and its compact form factor. It is a single-board computer which is affordable but has sufficient processing power for complex calculations, and, despite its diminutive size, it has capabilities that are comparable to desktop computers, making it ideal for embedded systems.

In this project, data is transmitted to the Raspberry Pi from the sensor nodes via LoRa. The collected data is stored locally in CSV files before it is processed by the Python scripts and passed to the machine learning model.

Edge computing, which refers to processing data locally rather than relying on the cloud for computation, is another key aspect handled by the Raspberry Pi, reducing latency and improving the accuracy of predictions, as well as eliminating reliance on internet connectivity.

The Raspberry Pi also plays a crucial role in communicating with other components in the system, such as the GSM module and cloud platforms, sending data to ThingSpeak for visualization, and initiating SMS alerts.

Overall, the Raspberry Pi is the core of the system as it provides a capable and flexible means for data processing, prediction, and communication at an affordable price.

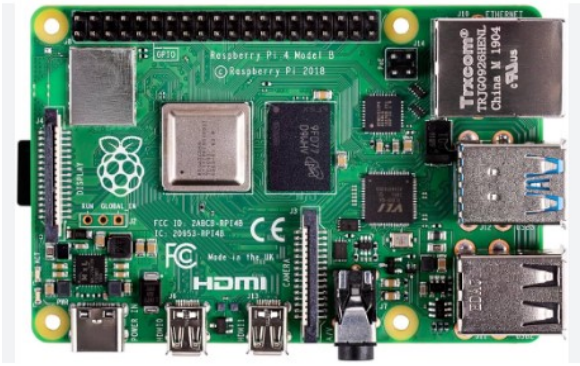


Fig 3.3 Raspberry Pi

ThingSpeak

For visualizing sensor data in a user-friendly and meaningful manner, we employ the cloud platform ThingSpeak. It's important not only to collect data from sensors, but also to be able to display it, interpret and process it. This is where ThingSpeak comes in.

In essence, ThingSpeak works like a dashboard where the environmental data gathered from all the sensors-namely water level, soil moisture, temperature, humidity, etc.- are displayed in graphs. Environmental variations can be easily interpreted using these charts, and in contrast to viewing numerical values, it provides a clear visualization of patterns, variations, trends, and fluctuations which helps in making right decisions.

In this project, by means of an internet connection, selected sensor readings are uploaded to ThingSpeak. Not all data requires constant upload, hence selected parameters were transmitted so that unnecessary load is decreased. Once the data reached ThingSpeak, it is stored into what is called a 'channel' where each 'channel' contains a variety of data 'fields'. Data 'fields' can then be plotted in real time to be used by anyone to monitor the environment around.

The advantage of ThingSpeak is its ability to display data in real time. This ensures that once data is sent from a particular system, it is instantaneously visualized on the graphs of ThingSpeak which enables the user to instantaneously notice the variations and trends. This is of extreme importance in a flood detection system where environment parameters can vary extremely fast, for the user/farmer to get immediate feedback.

Another important aspect is the ability to remotely access data through a smartphone, tablet, laptop etc by using an internet connection which ensures it can be monitored from any location in the world. This enables high flexibility as user can check his environment conditions while travelling, sitting in his office or relaxing at home.

Apart from visualization, ThingSpeak supports data analysis using features like integration with MATLAB which is highly suitable for performing more advanced data processing. While this project relies mainly on the visualization part of the ThingSpeak platform, it can be adapted for use in analyzing further data if necessary in future to expand the capability of the system.

It can be said that ThingSpeak has been one of the major contributors to making the system as user friendly as it is. The graphs are easy to understand and interpret and there is no prior technical expertise required. Therefore, it makes the system suitable for farmers or even people not technically savvy to use the system and monitor the environment efficiently.

Another important feature to be considered here is security. A public or private channel can be maintained and one may choose the accessibility to the data stored in a specific channel.

To sum up, ThingSpeak makes the system user-friendly and more interactive by visualizing data to be more understandable and accessible via a remote interface for effective and real time environmental monitoring and also acts as a "window" to monitor the whole system working in this case to detect a flood.

LoRa Module

The LoRa (Long Range) module is very crucial as it is responsible for the wireless transmission of sensor node data to the gateway. It is important in a scenario such as agriculture field where large areas of field need to be covered and internet accessibility is very limited. Hence LoRa is chosen.

LoRa is primarily known for its long range and low power data communication. Unlike other communication modules like Wi-Fi and Bluetooth where the distance is small, a LoRa module can transmit the data up to few kilometers but it uses very less power which make it an appropriate option for this kind of application. In this system, different sensor readings collected by the Arduino are sent to the LoRa transmitter module and then transmitted wirelessly to the LoRa receiver module connected to the Raspberry Pi. The whole transmission can be carried out without any need of the internet.

Low power utilization of LoRa is a very important advantage as sensors are battery-operated or solar powered which means very less power is utilized which also implies less maintenance.

The long range aspect of LoRa is again another major benefit since the distances between the sensor nodes to the gateway can be a couple of kilometers depending on obstacles in between, this also gives scalability to add more number of sensor nodes.

High data transmission reliability due to the use of spread spectrum technique used by the LoRa module makes it robust to interference and ensures reliable communication even in difficult circumstances. Obstacles such as trees, buildings or other geographical features might cause disturbances in communication, which will be suppressed by the LoRa module.

Furthermore, LoRa is one of the radio frequencies which operate in unlicensed bands; hence there is no expense involved for infrastructure and usage.

To conclude, the LoRa module is an integral part of the system to ensure long-range and reliable communication of sensor data between remote nodes and the gateway. It is a valuable choice as far as rural or agriculture usage is concerned.

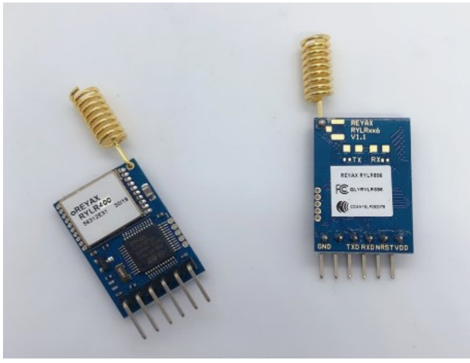


Fig 3.4 LoRa Module

GSM Module

This project makes use of a GSM module to effectively send alerts via SMS to warn of flood possibilities. For flood prediction, communication is paramount, and with SMS, timely updates can be made to prevent damage and ensure users are notified before the predicted flood has actually taken occurred.

In this project the GSM module is interfaced with the Raspberry Pi, which then sends an instruction to the GSM module if the machine learning model predicts a potential flood. The GSM module is then able to immediately send out an SMS to the user notifying them of the impending flood situation.

One of the most compelling reasons for utilizing a GSM module in a flood prediction system is its coverage. Most rural areas now possess mobile phone reception and as a result SMSing can now effectively relay important information even without internet access. Moreover, while not all modern phones have internet access and data subscriptions, most basic mobile phones can still receive and read SMS messages.

Another key reason is the ease of use and simplicity of the module, not only the hardware and connecting, but also the ease of programming it in Python. This system sends messages via SMS at a rate that ensures alerts are sent quickly to the designated recipients.

In summary, the GSM module is critical in the communication aspect of this project; providing an easy, reliable and universally accessible means of communication with the user of this system.

Fig 3.5 GSM Module

Scikit-learn

Scikit-learn is the library that has been chosen to facilitate the machine learning for this project, more specifically, the generation of the Random Forest classifier. It possesses all of the required functions to create, manipulate, and train a classification model. Scikit-learn is, in itself, a robust collection of algorithms designed for data analysis and prediction.

For this system, scikit-learn is used to take the input data from the sensors and create the machine learning model. This then allows the prediction as to whether or not conditions favor a flood.

A primary strength of scikit-learn is its ease of use and simplicity when writing code for a classification model, a multitude of readily usable functions for training and testing exist which help to streamline the process greatly.

Furthermore scikit-learn has been written with performance in mind, this will allow the model to predict using live data quickly enough to provide near real-time predictions of flood situations, even when the amount of data increases.

In conclusion, Scikit-learn serves as an indispensable tool for the classification part of the flood prediction system, and has been crucial to achieving reliable results.

CSV Files

CSV files are utilized within this project for the simple and efficient storage of all collected data from the sensors. Think of these files as basic databases. Each row in a CSV file contains information regarding a single set of sensor readings, with each column representing one of the measured parameters e.g. Water level, temperature etc.

These files are easily read and analyzed. Their structure makes it simple to extract and prepare the data so it can be fed to the machine learning algorithm; they can also be processed quickly and easily by programming language such as python and read in an application such as Microsoft Excel.

The CSV files used here provide a convenient format to store data which can later be used to build and improve the machine learning model.

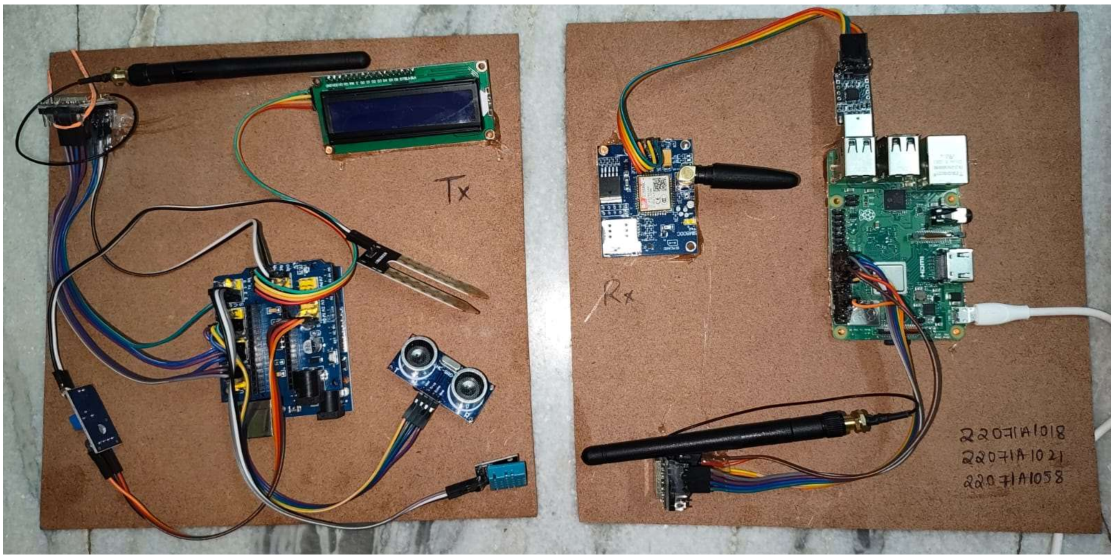


Fig 3.6 Circuit diagram of project

3.7 Justification of Selected Approach

The choice of a suitable approach for the design of an engineering system is one of the factors that determines the success of the engineering system as a whole. In the case of flood prediction where there are required accuracy, reliability, timely prediction and operational practicability, the chosen approach of this project of integrating IoT with machine learning algorithms has been made after considering its effectiveness and limitations compared to the already implemented solutions.

Reason 1: Need of the Hour The implemented system goes beyond simple observation and uses a predictive approach, rather than a reactive one which is found in most conventional systems. Most existing systems monitor whether water level crosses a certain threshold or some basic parameter like rainfall level. However, they lack predictive capability in such cases. Hence, by integrating IoT with machine learning, the proposed system can anticipate floods before they occur, and farmers can take measures to minimize damage to the crops and other agricultural structures.

Reason 2: Smart Monitoring and IoT The integration of IoT is essential to monitor various parameters like water level in the river, soil moisture level, air temperature and humidity. Internet of Things allows for constant monitoring of such data without any manual intervention. Also, by integrating various IoT devices, information from multiple sensors such as water level sensor, soil moisture sensors etc. Can be combined which would result in a better understanding and processing of environmental conditions that leads to the occurrence of floods.

Reason 3: Accuracy using Machine Learning Machine learning is incorporated into the system to analyze various collected parameters and detect patterns which are responsible for floods and thus helps in accurate predictions. There are several machine learning algorithms and Random Forest Classifier has been selected for this project based on its characteristics like robustness, high accuracy and effectiveness even with a dataset which may not be extremely large, good performance for imbalanced data (since floods do not occur daily). Random forest classifier builds multiple decision trees and merges their outputs. The randomness of samples helps reduce the correlation among them which increases accuracy. Random forest classifier gives more accurate and better predicted results than decision tree classifier and it avoids the over-fitting problem.

Reason 4: Low-power Long-range Communication and Data Transmission It is important to send the collected data from sensor devices in the agricultural field to the central processing unit. Traditionally WiFi and GSM technologies are used for communication. However, WiFi has a short communication range and it consumes a high power which is not feasible for distant fields and LoRa communication has been utilized in this project. Long Range LoRa (Long Range) technology allows communication across long distances with reduced power consumption and with no continuous internet connection needed. Hence it is ideal to transmit data from various locations to the main server.

Reason 5: Effective alerting system An efficient alert system for notification to farmers regarding predicted floods has been an essential part of this project. It has been implemented using GSM module and it sends an SMS alert directly to the farmer's mobile number, thereby facilitating them to take preventive measures even in regions without internet facilities.

Reason 6: Automated Process The entire system, from data collection to analysis and prediction to notification of alerts, is automated. This automation reduces the involvement of humans in monitoring and processing and hence the process is efficient and fast, which is a requirement in case of floods.

Reason 7: Scalability The system is designed to be easily scalable by adding more sensor devices and other components in future as and when required, without much of an investment. The system can be extended for monitoring and prediction in larger agricultural fields and regions.

Reason 8: Cost-Effectiveness The system uses inexpensive components like Raspberry Pi and Arduino boards, various readily available sensors and open-source software. LoRa modules also are relatively inexpensive compared to alternatives like cellular modules for rural areas. Thus the whole system is cost-effective and suitable for widespread deployment in agriculture sector especially in developing countries.

The chosen approach is a significant improvement over existing solutions that mainly focus on observation or use a limited number of parameters. It offers higher accuracy, better timeliness, enhanced automation, and cost-effectiveness. By integrating multiple technologies, it addresses the need for a comprehensive flood prediction and alert system for agricultural applications. The independence from internet connectivity in the field makes it practical and reliable for remote areas. In essence, this approach provides a holistic and efficient solution for a persistent challenge in agriculture.

CHAPTER 4

DEVELOPMENT AND IMPLEMENTATION

4.1 Implementation Plan and Phase–II Progress

Phase-II of the project marked the shift of efforts from designing and analyzing the theoretical framework to the actual implementation of the proposed IoT based flood prediction system. This phase was important as it led the theoretical designs and ideas into tangible prototypes. A structured and systematic approach was followed while developing and testing the different modules. The process involved the implementation of both hardware and software aspects of the system concurrently to create a fully operational product.

The first step in Phase-II was the selection and purchase of all the required hardware components. This involved obtaining sensors such as ultrasonic sensors for detecting the water levels, soil moisture sensors for determining the saturation level of the soil and DHT11 sensors for monitoring the temperature and humidity. In addition to the sensors, the various components of the system like Arduino microcontroller, Raspberry Pi, LoRa communication modules, GSM module, connection wires, power supplies, and display devices were bought. Efforts were made to obtain components that are economical and dependable, and suitable for agricultural conditions.

After procuring the hardware, the actual implementation of the system was undertaken in stages. By adopting this strategy, the different modules of the system could be tested in isolation before being integrated with other modules. This aided in the detection of issues at an early stage.

Initially, all the sensors were tested independently by connecting them to the Arduino microcontroller and writing individual Arduino sketches. Ultrasonic sensors were tested for distance measurement (to determine water level). Soil moisture sensors were tested for moisture reading. Similarly DHT11 sensor was tested for measuring temperature and humidity. During the testing phase, accuracy and calibration of sensors were examined and checked, and Serial communication was used for monitoring the readings of the sensors at the time of debugging.

Following the tests on the individual sensors, all the sensors were combined into one module by integrating all the sensors to a single Arduino board. Programs were written using multiple programs to collect readings from all the sensors together at specific time intervals. Care was taken to see that individual sensor readings do not affect each other.

After successfully integrating and testing the sensors with the Arduino board, establishing a communication link between the sensor nodes and the gateway node became a task. This was done using LoRa communication modules, one placed in the sensor node and the other in the gateway node (Raspberry Pi). Data packets were configured for transmission and reception. This helped the system to communicate over long distances without internet, thereby reducing the operational costs.

The software aspects of the system were concurrently developed on the Raspberry Pi using Python due to its wide library support and simplicity. Various programs were written such as for the data acquisition, logging of the data, for the machine learning algorithm to predict potential flood occurrence, for real-time monitoring of the system, and also for alerting the user in case of any potential threat.

The data logging module stores all the collected sensor readings into a structured format such as a CSV file, in order to save the data that was collected at different times for later analysis. The data logging module was made efficient in order to maintain the continuity of data transfer.

A machine learning algorithm, Random Forest, was built to classify whether an impending flood condition exists or not based on the different readings from the sensors and using the available data set. This training was carried out by providing data of various conditions and the corresponding output indicating whether or not a flood is possible was given to the algorithm so as to enable it to classify and learn the patterns. The developed machine learning model was deployed on the Raspberry Pi for processing the incoming data.

The real-time monitoring application displays the current data obtained from the sensors along with their corresponding interpretation (e.g. Normal, high level, critically high level). Real-time system monitoring provides an interactive and intuitive display of system information and is essential to know what is happening at each moment in the agricultural field.

The SMS alerting system was integrated with the system using the GSM module. This makes the system more user-friendly by directly alerting the concerned people by sending an SMS about any potential flood condition that could occur so that prompt actions can be taken.

The complete system was then tested end to end by simulating different kinds of conditions and ensuring that it performed according to the predictions that were set. A series of tests were performed to ensure its accuracy and reliability in collecting, processing, transmitting, and acting on the data to send notifications at the desired moment.

Many difficulties were faced during phase-II but each were rectified at different stages of the process. For example, calibrating the sensors proved to be difficult, establishing the communication between the sensor nodes and the gateway was also a complicated task, but ultimately all challenges were overcome due to a proper systematic approach to development and implementation.

In conclusion phase-II of the project can be said as the successful phase where the theoretical concepts developed in phase-I are implemented as functional prototypes, thereby paving way for the real world applications of the proposed IoT based flood prediction system.

4.2 Development / Execution of Major Components

The success of the proposed flood prediction system depends on the development and execution of the two major components, i.e., the hardware components and the software components. Both of these components work together to realize the real time monitoring, data processing, prediction and alarming functions. Two major subsystems of this system were developed, which are described below:

Hardware Development - Sensor Node

This node is basically a front-end part and performs the task of data collection from field and sending it to gateway node. The sensor node was developed using an Arduino microcontroller as its central processing unit. This node comprises of multiple sensors, water level sensor, soil moisture sensor and temperature and humidity sensor, are used. These sensors are deployed in the agriculture field to read actual conditions of different environment parameter relevant to flood prediction.

The signals from the various sensors, which may be analog or digital, were then captured and converted to appropriate meaningful values using an Arduino and programmed using Arduino IDE. Care was taken regarding the synchronization of readings from the various sensors to avoid any inconsistencies.

Sensor calibration was considered as the crucial part in the hardware development, so that the output of each sensor could be reliable. Various parameters were used to calibrate the sensors under different conditions. If the sensor reading is not correct then the performance of the machine learning model may be affected and incorrect prediction of the risk of flooding may be produced.

The data collected was sent to gateway node using a LoRa radio frequency transceiver module with an Arduino which acts as an adapter to send the data. This is good for low power consumption and for remote communication.

An LCD was also incorporated into the sensor node for real time viewing of the data without any need for other apparatus. This made the system easier to use.

Hardware Development - Gateway Node

The gateway node acted as a brain of the whole system. This node is made using a Raspberry Pi. It receives the data from the sensor node using LoRa module receiver, and processes and analyzes the received data. The Raspberry Pi was used because it is inexpensive, small in size and powerful enough to handle machine learning algorithms.

Initially, it was necessary to set up the software in Raspberry Pi. So, the Operating System was installed, then python programming was enabled along with installing all the necessary libraries required to carry out different functions.

Software Development - Data Handling and Processing

Python was used for programming to handle the incoming data from the LoRa module. The received data were decoded and stored in a comma separated file (CSV).

Each row of the CSV file represents one reading taken from each of the sensors and parameters like water level, soil moisture, temperature and humidity have each been attributed a separate column to easily extract the values and perform machine learning.

Machine Learning Implementation

The machine learning model used for predicting the risk of flood was based on the random forest algorithm implemented in the Scikit-learn library. This was done as it produces higher accuracy for the dataset. The random forest model takes the data collected from the sensors and predicts whether there is an impending risk of flood.

The machine learning model was trained on data that was collected in the earlier part of the project. After training, this model was deployed in the Raspberry Pi which receives new readings of the various sensor values and produces an outcome.

Alerting Mechanism - GSM Module

The GSM module was incorporated so as to provide SMS alerting to the user of potential flood risk. This would be especially helpful in rural areas that may have a lack of internet access. The communication was performed using AT commands and the readings were stored. When the model produces a prediction stating there is an imminent risk of flooding the system would automatically sends out an SMS to notify.

Web-Based Application

In addition to the GSM alerts, a web based application was developed where all the data could be displayed on screen as to show what is going on with the environment. This made it easier for the user to quickly see if there was any risk of flooding.

Overall Integration

All of the components were programmed so that they work efficiently together. The sensor node transmits data, the gateway node processes this data and sends it to the web based application for the user, and in cases of potential floods, the SMS alerts are sent out by the gateway node as well.

In summary, the main hardware and software components that constitute this flood prediction system were developed. These were developed and successfully integrated so as to realize the function of prediction and alerting of flood risk to help out the people who might be prone to flood risk in the agriculture field.

4.3 Integration and Overall Functioning

A successful implementation of the proposed IoT-based flood prediction system heavily relies on the seamless integration of all the hardware and software modules. After individual implementation and testing of the components such as sensors, Arduino, LoRa communication module, Raspberry Pi, ML model, dashboard, and GSM module; these were integrated together to form a complete end-to-end system. Through this integration, the system is able to act as a one complete system to monitor the conditions, predict the risks of floods and alarm users with the warnings.

The system operates in a continuous and cyclic process whereby the data passes from the sensing layer to the end user, in stages. The system performs this operation in an endless loop whereby the system monitors the surrounding conditions. Due to the integration of various components, a smooth flow of data between them is guaranteed.

Initially, the various sensors present in the field such as water level sensors, soil moisture sensors and temperature and humidity sensors read the respective data values. These sensors provide the readings that are required to detect any flood-prone condition in the surrounding.

After reading all the sensor data values, they are transferred to the Arduino microcontroller, which acts as the primary data acquiring unit. The Arduino microcontroller reads the various sensor readings in the digital format, processes and transfers it to the communication module after conversion to serial format. This continuously process keeps on on for some particular amount of time for each of the sensor.

After reading all the sensors and transferring the data in a serial format the data is transferred to the communication module (LoRa transmitter) of the system for wireless data transfer. The data are transmitted using a LoRa transmitter at long distance with low power and the data is received by the LoRa receiver which is interfaced to Raspberry pi.

In the communication module (gateway) phase, the data received from the LoRa receiver is transferred to the Raspberry Pi which acts as the control unit. The Raspberry Pi performs many functions as mentioned below:

(1) Stores all the data values received in a CSV file. The file is saved on the Raspberry Pi storage where it can be accessed whenever required.

(2) Analyzes all the collected data through Python script and inputs it to the ML model to determine the risk of floods. The Random Forest algorithm is used to analyze data from various sensors and determine if any Flood Risk exists.

(3) Updates the data to be presented to the user at the dashboard, where the graph of various sensor data and status are updated continuously.

(4) When the ML model determines that there is a risk of Flood, the command is sent to the GSM module to send an alert SMS. The GSM module sends a text message alert to the users that there might be a flood risk ahead.

The entire system operates in a loop with each component taking input from the previous stage and producing output for the next stage in a seamless way. With the integrated system the system remains active all time and responds in rapid time without any time delay to changing environmental conditions. With the usage of different communication technologies, the overall robustness of the system is maintained even if some fault occurs. For example, if there is problem with the internet then, the alert could be still sent using SMS via Gsm module. Moreover, the proposed system can be scaled or enhanced in future by the addition of additional features such as further more sensors or machine learning models to analyze data in various patterns for an improved accuracy of flood detection.

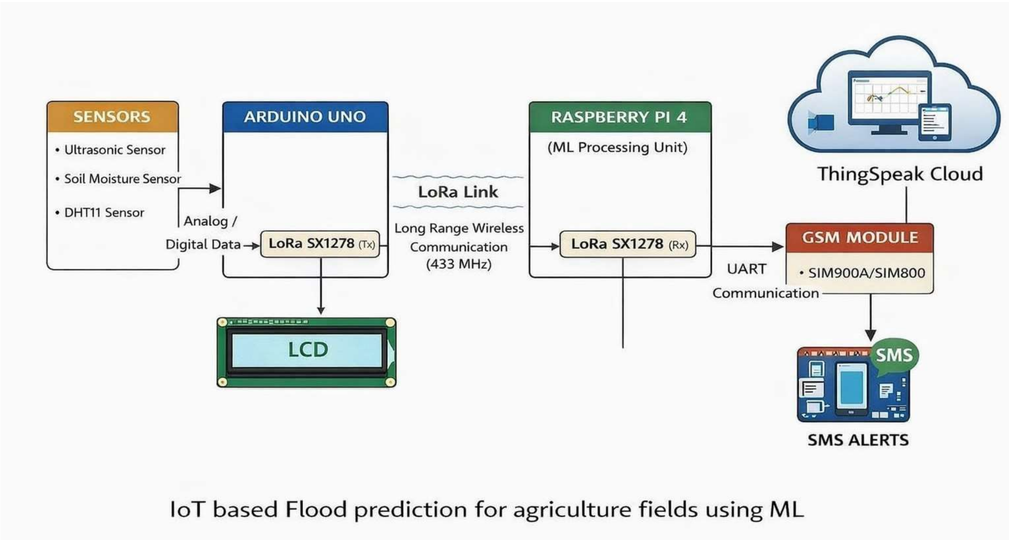


Fig 4.1 Block Digram

4.4 Challenges Encountered and Solutions

Throughout the implementation of the IoT-based flood prediction system, there were several real-world issues encountered across both the hardware and the software. Many of these were issues common with real-time system development and especially for multi technology based system development that combines different components such as sensors, communication modules, embedded systems and machine learning, although these were carefully investigated and addressed using systematic troubleshooting and optimization to improve the system performance and robustness.

One of the major issues faced during the implementation stage was related to the inconsistency and instability of data collected by the sensors. The sensors, namely water level sensors, soil moisture sensors and temperature & humidity sensors at the beginning produced unstable output. These inconsistencies were mainly due to issues with calibration, environmental interference, and unwanted signal interference from external parameters such as varying temperature, ambient moisture and electrical disturbance. These were crucial issues since the data input directly influences the performance of the machine learning algorithm.

To combat these issues, calibration of each sensor was performed, where each sensor was carefully tested and adjusted for it to produce consistent readings of expected values. Simple filtering methods such as averaging of readings and threshold based filtering were also implemented to increase the stability of sensor readings. Such changes ensured the data being processed by the algorithm was accurate and reliable.

Another major problem was experienced with the LoRa communication system. Even though LoRa allows for long range communication, there was inconsistency in data transmission such as loss of packets and communication interruption. This was primarily due to environmental obstructions such as objects acting as interferences to the transmission, incorrect positioning of the antenna and improper transmission parameters. This caused the complete system to not function in real time and affected the machine learning algorithm’s predictions.

To overcome this problem several optimizations were implemented. The antenna positioning was made as optimal as possible by trying different locations and using efficient antennas and also by positioning the sender (Arduino with LoRa module) and the receiver (Raspberry Pi with LoRa module) carefully. Also, the transmission parameters such as frequency, spread factor and transmission power were adjusted to establish reliable and stable long distance communication with minimal packet loss.

The third problem was faced with the software integration aspect; particularly with file operations and data synchronization between the different modules of the software. Since there are multiple processes like receiving data, writing it into a CSV file, processing the data, and eventually sending out alerts, ensuring proper synchronization and avoid conflicts was essential. During this phase, lagging data input and conflicts when accessing the file were some of the problems which affected the whole system performance.

To address this, careful software re-architecture and optimization was carried out. A proper data synchronization technique was implemented to ensure efficient transfer of data between the different modules of the system. Efficient file handling techniques were employed in order to secure the file and prevent access conflicts, and also efficient coding was applied to allow faster writing and reading. Error checking and handling was also incorporated to help catch errors easily.

The fourth major problem was encountered with the GSM module where it had communication problems between the Raspberry Pi and itself. The GSM module did not work properly and it was unable to transmit any messages. This was due to failure in establishing connection to the network, issues with sending AT commands and also many communication errors between the modules.

To fix the issues, thorough debugging of the module's commands was performed. Proper initialization of the GSM module was ensure and AT commands for activating the module were carefully set up. The connectivity check for registering onto the network was added and only when the network was successfully registered was an SMS transmitted.

Other minor issues such as fluctuation in power supply, wiring connection problems and compatibility issues with different components were also encountered during system development, which were resolved by conducting tests on the components.

The numerous problems encountered during system development provided a lot of learning opportunities to fine-tune and optimize the system. By systematically troubleshooting and rectifying errors that occurred, the system was made to perform as expected. This optimization process made the entire system more reliable and efficient.

In conclusion, there were several technical difficulties experienced while building the system that were efficiently tackled with smart engineering practices and system design. These efforts led to the smooth performance of the entire flood prediction system with no compromises in the prediction accuracy or system functionality. The ease with which the problems were solved shows the practicality of the developed system for its intended use.

4.5 Observations

Several observations were noted during the testing and deployment phase of the proposed IoT based flood prediction system. These observations provided insight into the effectiveness and viability of the proposed system and helped to measure against the defined objectives of the previous chapters.

The ability to predict the flood situation effectively due to the incorporation of various sensor inputs is a key observation. Unlike existing flood prediction models that use single parameter like water level or rainfall for flood prediction, the system used various parameters like soil moisture, temperature, humidity and water level which provide the broader view of various environmental parameters affecting the floods. As, one may observe that even though there may be moderate rainfall, a higher level of soil moisture indicates a greater risk of flood and vice versa. Thus, with the incorporation of these various sensor parameters, the system is capable to predict the flood more accurately than the system using only one of the sensors.

Among various parameters incorporated in the system, the role played by soil moisture and rainfall were significant for the accurate prediction of flood. The soil moisture helped the system to calculate the absorption capacity of the soil. A higher absorption rate makes the soil saturated quickly and thus any further increase in rainfall causes flood. Similarly the rainfall was an important factor. The role played by both these parameters together leads to better flood prediction.

The Random Forest machine learning algorithm that was implemented into the system to carry out the data processing and prediction was highly efficient. One may observe from the various result sheets that the model processed data quickly and produced predictions despite having a small amount of data. Such a trait of the model would prove to be highly useful in agricultural sectors where the amount of data available may not always be large. It identified and used the various relations and correlations among different environmental factors to distinguish whether conditions are normal or prone to flooding.

The model gave highly accurate classification result, correctly classifying the environment as either safe or risky to flood. It also showed very low latency so as the prediction was always generated quickly as soon as the input from various sensors were received. Such speed of prediction will ensure efficient delivery of alert and warning to the end user for the prevention of flood.

The performance of communication system implemented was also observed in the testing phase. The LoRa based communication was found to be highly reliable for long range data transmission despite of the existence of various objects such as buildings or trees that would have attenuated the signal. Also, the low power utilization feature of the LoRa modules is highly beneficial in an agricultural application field, where access to power may not always be readily available. Another advantage observed using LoRa was the communication without the internet connectivity, unlike that required for Wi-Fi systems. This will be highly useful in the rural agricultural lands where internet coverage may not be always readily available.

The GSM based alerting system also gave effective result in the testing phase, since the alert notifications were received on time, once the prediction signaled that a flood situation is imminent. Moreover, GSM module has an advantage over internet based notification systems; as it will function irrespective of internet connectivity and will therefore be reliable in remote areas of agricultural applications, where mobile network access would be more easily available than internet access.

The system operated well as a whole as one integral unit, with all the various components of the system functioning smoothly in association with each other. Data from sensors were successfully read by the raspberry Pi and was passed on to the ML model for processing using LoRa. Finally, the classified information and the alerts were communicated to the end users using dashboards and via SMS. The loop in the system ensures continuous flow of data in a loop for immediate decision making and alerting.

The system proved to be quite extensible and flexible in terms of future scaling and modifications. More sensor nodes may be added for coverage of a larger area or for the inclusion of more sensor parameters may be added.

Also, the system was user-friendly and practical. The dashboard interface facilitated the easy monitoring of various environmental factors whereas the SMS alerting was extremely user-friendly to use even for those lacking technical expertise. The system is accessible to all level of users.

In general, the system is considered to be highly efficient and useful for the predicted task. Various deficiencies were addressed in the previously discussed existing flood detection and prediction system using the IoT and ML techniques combined with the smart communication and alerting techniques.

In conclusion, it was observed that the system fulfills all the desired objectives and works effectively in the real environment. The system showed the potential to perform very well if deployed widely and hence proved to be an efficient and reliable system for flood monitoring and predicting in agriculture domains.

CHAPTER 5

RESULTS, ANALYSIS AND VALIDATION

5.1 Testing / Evaluation Methodology

The testing/evaluation of the proposed IoT based flood prediction system was conducted through a combination of rigorous controlled testing and the simulation of real world conditions. The objective was to determine the accuracy, reliability, response rate and overall effectiveness of the system under a variety of different environmental conditions. Varying these conditions allowed for a broad understanding of the system's capabilities to effectively respond to real-world changes.

The test method implemented was designed to best replicate an agricultural environment, and since flood events are dictated by various environmental factors, important elements of this include the water level and soil moisture readings as well as the surrounding environment and general climate. By testing varying conditions within these ranges a picture of the effectiveness of the system against changes was formed.

Testing begins at the sensor level, collecting environmental readings from the aforementioned sensors which constantly record and transmit their information to the Arduino, which prepares them for communication to the Raspberry Pi. It is important that all collected data is accurate, as it is critical to the performance of the overall system.

Once collected the data is transmitted to the Raspberry Pi through LoRa communication. The LoRa protocol was chosen as it is a long range wireless communication protocol allowing for the sending of data without the need for internet access. The data collected is then stored on the Raspberry Pi in CSV (Comma separated values) format where it is then ready to be used for the machine learning aspect of the project.

This data is fed into the machine learning classifier in the system. The Random Forest classifier was chosen as it is able to handle the multiple variables collected from sensors at the sensor nodes. The dataset created is then split into a training set, and a validation set.

The training set is used to train the machine learning classifier to learn from the inputted data about normal and flood conditions. The validation set is then used to test the machine learning classifier against the trained data that has not been fed into it before in order to establish accuracy of detection.

Several tests were designed to be carried out. These included both normal conditions and conditions that would cause an event of flooding. Under normal conditions the system is not affected by external factors to a great extent and a correct reading (no alert) would be detected by the system. In flood conditions water level readings would be abnormally high with possibly a constant or increase in the readings due to heavy rain, soil moisture readings will be saturated, the climate sensors will show high temperatures, and a high level of humidity, therefore the system is expected to display the flooding state, and trigger alerts.

The tests run to observe the response rate and accuracy showed that with a change in water level, the system would respond almost instantly with an alert, once the machine learning model has correctly classified the inputs it was given, the output of the system accurately reflects this and identifies whether the conditions are at risk of flooding or not.

The overall performance of the system under these varied conditions was evaluated and found to be reliable. Communication issues were minimal using LoRa even over a long distance, the alerts were successfully delivered by the GSM module and the overall speed was excellent due to the processing capability of the Raspberry Pi and the simplicity of the chosen machine learning algorithm. The system also proved stable and could be run for a lengthy duration without faltering.

In summary, the implemented testing method has proved the effectiveness of the IoT-based flood prediction system, and shown its capabilities to deal with the challenges encountered within its application, and successfully implement the proposed solution to a real world problem.

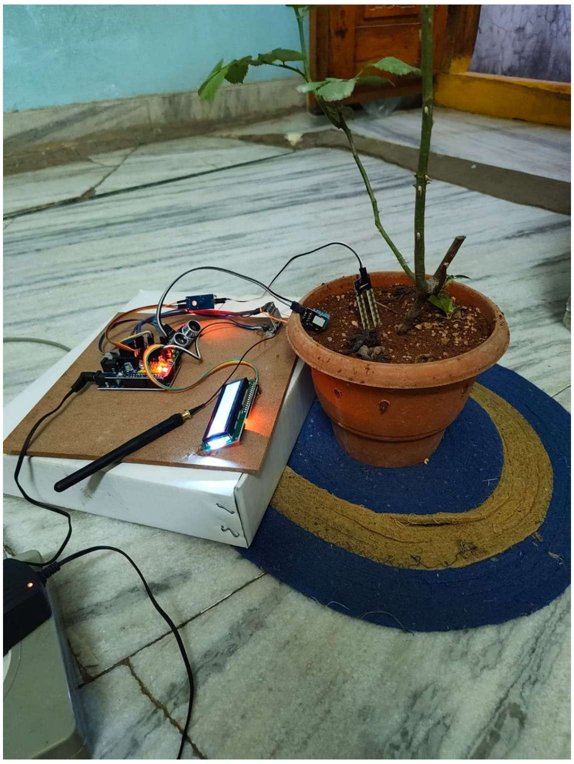


Fig 5.1 Testing on flower pot (indoor)

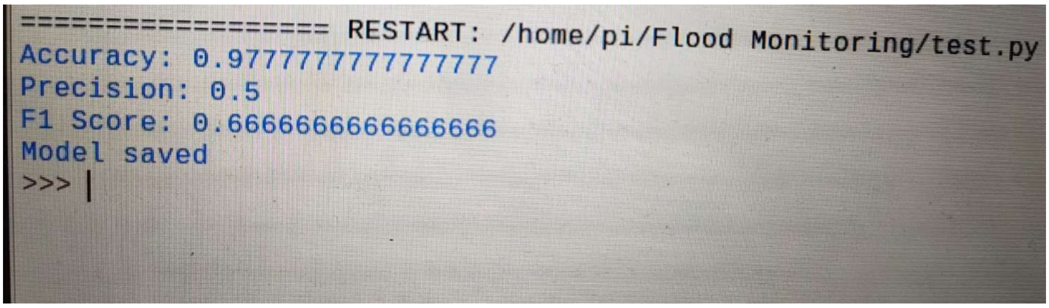


Fig 5.2 Acccuracy

5.2 Results Obtained

The results generated during the testing and implementation of the developed IoT-based flood prediction system proved to be accurate, reliable and efficient in agricultural real world application. The system was evaluated for both normal conditions and flood prone conditions in varying environments to analyze the prediction accuracy, response time, communication reliability and overall system efficiency.

Prediction accuracy was one of the key performances of the system. Machine learning model (Random Forest classifier) trained using a dataset collected from sensor nodes, including water level, soil moisture, temperature and humidity parameters, was evaluated on new unseen data. The analysis revealed that the model can accurately classify the normal and flood prone conditions. Moreover, employing multi parameter classification system helps the model identify and learn complex patterns which provides more reliable results as compared to single parameter prediction systems.

The system also performed favorably in terms of generalization. This characteristic shows that the model will also work efficiently even with unseen data and different environmental conditions and will not require frequent retraining. The Random Forest algorithm is particularly suitable for modeling non-linear relationships among variables, which can arise when modeling flood condition.

Another important performance criterion for the system is response time. The system is built for real-time monitoring and control which enables it collect and analyze data without interruption. Edge computing has been implemented to the raspberry Pi board that handles all data processing and prediction locally and makes it independent of any other external server. This ensures minimal latency in the system and quick prediction. Test was conducted and found to deliver timely updates in predictions which is vital for this type of time sensitive application.

Communication performance was another critical factor tested. LoRa technology made it possible to establish long-range communication, system was tested at different conditions and found to provide very robust communication. Data was transmitted from the sensor nodes to the gateway with minimum packet loss.

Furthermore, low power LoRa modules makes the system very efficient in terms of battery power and ideal for long-term applications in remote area.

Data storage and handling is another factor analyzed. Storing sensor readings as CSV files proved to be very convenient and efficient for further use and processing by machine learning model. The system handled continuous stream of incoming data without any issue.

Alerting mechanism using GSM module for sending SMS was also tested and found to be very responsive. Timely messages about the potential flood threat were successfully delivered to the users in rural environment where there is limited access to the internet. The system proved to be efficient to respond within time and ensure proper alerting of users in event of emergency.

Dashboard and visualization display was found to be clear and user-friendly. Users can monitor the environmental data and observe the trends which helps in proper decision making.

It was also noticed that the system could sustain its operations for a considerable duration without any failure or shutdown. It can also be scaled to work with a larger network by adding more nodes to cover a wider area. Future enhancements such as integration with weather forecasts can be implemented in future to further increase the accuracy.

The results confirm that the developed IoT based system is an effective and reliable system which would provide the farmers with timely alerts, hence assisting them to minimize the risk of flood damage and achieve greater yields, leading to increased sustainability of agriculture.

	Time	Temperature	Humidity	Water_Level	Soil	Flood
143	2026-04-08 18:48:51	30.6	34.0	15.0	1023.0	0.0
144	2026-04-08 18:49:27	30.6	34.0	17.0	1023.0	0.0
145	2026-04-08 18:49:49	30.5	34.0	16.0	1023.0	0.0
146	2026-04-08 18:49:56	30.6	34.0	14.0	1023.0	0.0
147	2026-04-08 18:50:10	30.5	34.0	10.0	1023.0	0.0
148	2026-04-08 18:50:17	30.5	34.0	11.0	1023.0	0.0
149	2026-04-08 18:51:08	30.5	34.0	11.0	1023.0	0.0
150	2026-04-08 18:51:51	30.5	34.0	13.0	1023.0	0.0
151	2026-04-08 18:52:19	30.5	34.0	13.0	1023.0	0.0
152	2026-04-08 18:52:48	30.5	34.0	13.0	1023.0	0.0

Fig 5.3 – Sample Results

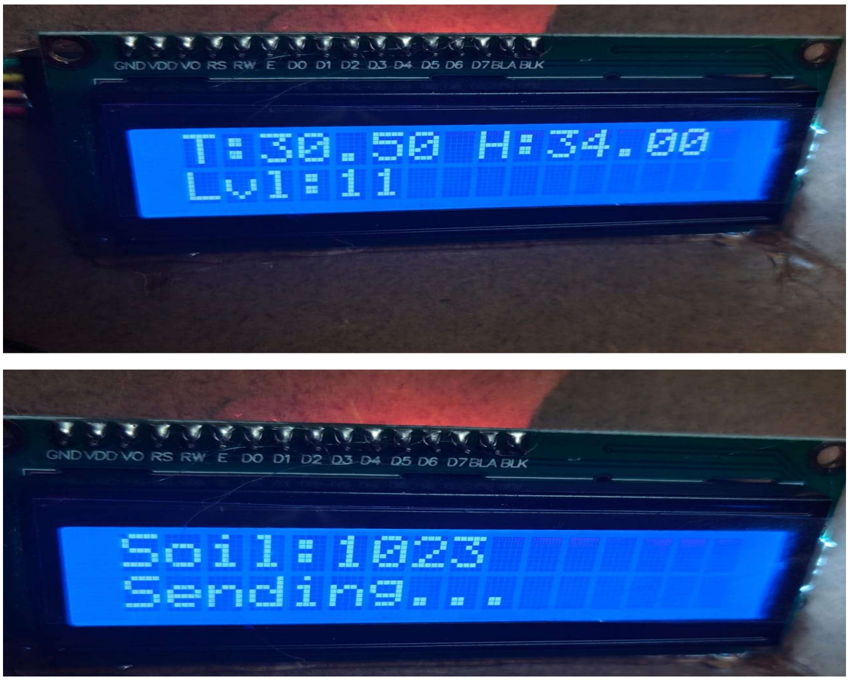


Fig 5.4 LCD Monitoring of Sensor samples

5.3 Analysis and Interpretation of Results

Analysis and interpretation of the results obtained from the testing phase are important in understanding the effectiveness and reliability of the proposed IoT-based flood prediction system. The examination of the outputs from the system such as the predicted results, performance matrices, and data visualizations helped us in evaluating how well the system is performing under various conditions, and its accuracy in detecting potentially flooded conditions.

One of the significant points concluded from the analysis of results is that the machine learning model exhibits a high degree of accuracy and effectiveness in predicting floods using several environmental parameters. It is not just dependent on one input factor but uses a combination of multiple parameters such as water level, rainfall intensity, soil moisture, temperature and humidity. The multi-parameter approach helps in significantly increasing the predictive capability of the system. It was observed that using only water level to predict floods would not be enough, as even though water level provides direct input about the rising levels, it does not explain the cause.

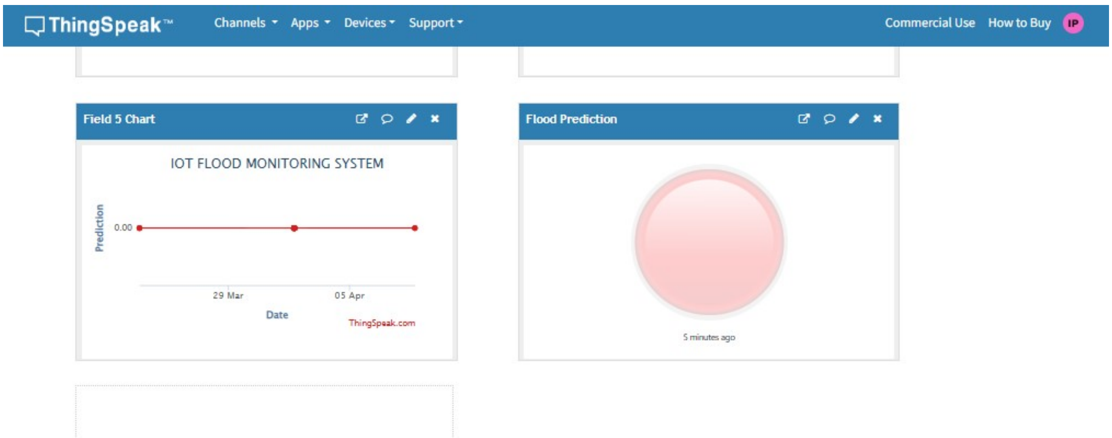
Rainfall intensity and soil moisture play a vital role in predicting floods more accurately. Rainfall intensity gives a direct input about the amount of rain falling, while soil moisture is a parameter which states the saturation level of the soil and in case the soil is saturated then a moderate amount of rain can be harmful, and the result was indeed obtained by combining the results of the two parameters, so one cannot use only one input.

The results indicate that the machine learning model being used (Random Forest classifier) is a very effective classifier for categorizing the inputs into normal or flooded conditions. Random forest classifier is an ensemble machine learning algorithm, which makes prediction by fitting a number of decision trees to the sub-samples of the data and taking their average or mode for prediction. So with these properties it performs well in fitting the complex relations among different environmental parameters.

The model is evaluated based on accuracy, precision, recall, and f1-score and the results obtained were very high in terms of the values of accuracy, precision, recall, and f1-score for the normal as well as the flooded conditions. High accuracy shows that the overall predictions made by the model are correct. High precision indicates that the predictions of flood conditions are not that frequently wrong (false alarms), and high recall signifies that most of the actual flood conditions are detected correctly.

A balance of precision and recall must be maintained to minimize the prediction error in terms of both false positive and false negative which are demonstrated by lower values in some cases. Visualization of data through the dashboard and using platforms such as ThingSpeak provided the strong relationship between the parameters. It was found that after an increase in rainfall intensity, the soil moisture level is raised which leads to the water level to increase. Such trends are clearly observable from the graphs in a pictorial form, thus helping in better visualization of how they collectively affect.

The real time performance and the low latency achieved from the system also signifies its effectiveness for the practical applications where quick and timely results are needed. Integration of different system components was done smoothly, to ensure proper and coordinated function of each part and to contribute to the overall effectiveness of the solution. The system has strong potential for extensibility, and to incorporate more parameters and to increase the prediction accuracy using a large amount of data.



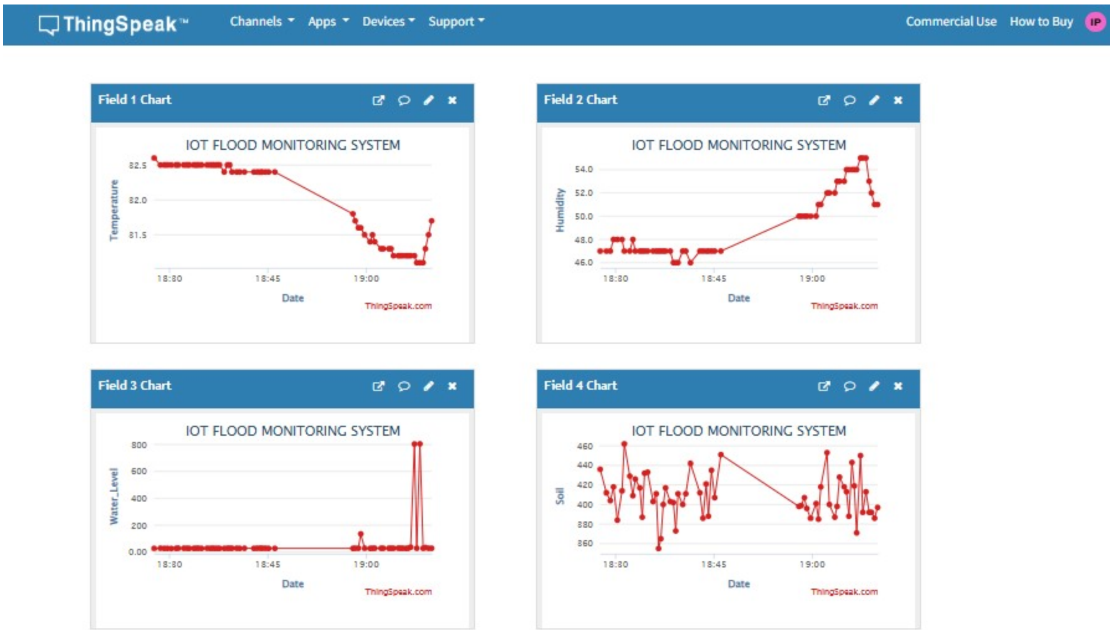


Fig 5.5 Graphs

5.4 Validation of Objectives

Validation of the objectives is a significant step in the evaluation of the developed IoT based flood prediction system. In this section, it is examined whether the objectives defined in Chapter 1 have been successfully accomplished using the implemented and tested system. Based on the findings obtained in the course of the development and testing phase, it can be observed that the proposed system has met the major objectives specified in the problem.

The first main objective of the system was to enable real-time monitoring of environment parameters using the capabilities of IoT devices. It can be observed from the results that this objective has been accomplished using a number of sensors, including the water level sensor, soil moisture sensor, temperature and humidity sensors. These sensors collect information about the parameters at an interval of time and pass it on to the gateway. The Arduino microcontroller performs certain operations on the information obtained from the sensors and the same is passed on to the gateway. This operation goes on continuously in the loop, resulting in an uninterrupted sensing of environmental parameters, which can be directly related to predicting floods.

The second main objective of the project was to set up an effective communication network for the transmitted data over a long range. This objective has been successfully accomplished using the LoRa technology (Long Range communication). The results show that the LoRa communication works efficiently with good connectivity between sensor and gateway. The amount of time taken for the data transmission is negligible. Another benefit achieved through LoRa communication is low power consumption. LoRa communication requires less power, and hence it is quite useful for application in rural and agriculture fields. The communication doesn't require internet to transmit data, thus proving more reliable even in an environment without network connectivity.

The third major objective of the system was to provide a predictive solution through machine learning for the prediction of floods. This objective has been accomplished by using the Random Forest algorithm with the help of the Scikit-learn framework. It is clearly seen from the results that the developed machine learning model performs efficiently for the detection of flood events and can predict flood events in a given condition. It takes all the possible parameters from sensors in an input to give output which would then tell about the possibility of flood occurring. This significantly helps in avoiding natural disaster and reduces loss of life.

Another important objective was to provide effective data visualization for the analyzed information of sensor readings to assist the user in readily interpreting it. The results of the implementation confirm that this has been successfully achieved through the web based interface integrated with ThingSpeak. It enables to clearly visualize the current and historical values of the environmental parameters in graphs and charts. The availability to retrieve the data from cloud platform ensures data visualization to be accessible and more user-friendly.

Furthermore, one of the critical objective was to implement an efficient alert system to notify users. This objective has been achieved by integrating a GSM module for sending SMS alerts. As the result shows that the system sends alerts directly on the mobile phones of the user when flood condition is detected. Even without internet connectivity, these messages can reach directly to the user, thus, it has a better performance in the remote area with the unavailability of internet. Timely alerts from this system saves the users from taking risky decisions.

Apart from the main objectives mentioned above, there are other minor objectives achieved such as cost-effectiveness, scalability and ease of implementation of the proposed solution. Use of low-cost sensors, microcontrollers such as Arduino, Raspberry Pi etc. Are economical as they are readily available. The use of open-source libraries and framework is another factor of cost-effectiveness. The scalability allows the system to cover more area and incorporate additional features and sensors if necessary.

In order to conclude it, it has been strongly justified that all the main objectives of the developed IoT based flood prediction system as specified in the chapter 1 have been achieved effectively due to its successful operation. All the functionalities provided by the different modules of the system i.e. Sensing, communication, processing, data visualization and alerting are working well together and giving appropriate results as per the requirement and hence the developed system is fully capable of predicting flood in an efficient way.

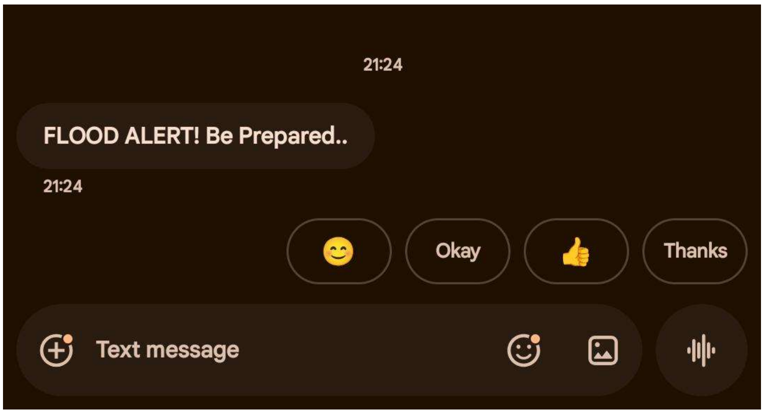


Fig 5.6 SMS notifications were delivered through the GSM module

5.5 Comparison with Expected / Existing Outcomes

When compared with existing conventional systems and IoT-based systems, the proposed system is found to be significantly improved in monitoring, prediction ability, accuracy, communication methods, alert systems and cost. Conventional monitoring systems mainly worked on human manual checking of water level and collection of periodic data, which was time consuming, not prone to errors and not real time. Existing IoT systems like proposed system use automatic sensors which monitor parameters and collect them without human monitoring. In proposed system various sensors like water level, soil moisture, temperature and humidity are used which provide more information for better monitoring.

The most crucial one in the above discussed parameters is prediction ability. Traditional systems do not provide any predicting features, but only display the past/current condition. Existing IoT based systems have poor predictionability based on threshold value or simple data analysis method. Existing systems are not accurate at the times of changing parameters, but the system is more accurate as it used machine learning based prediction algorithm; especially it used advance algorithms like Random forest to calculate potential risks for flood.

One more important factor is accuracy. Traditional system had less accuracy as it mainly worked on manual check. Existing IoT based systems had a little more accuracy as it used automatic sensors but it does not collect multi sensor data at a time. The system designed by us had more accuracy as it used multi sensors and also a machine learning based algorithm. With the use of multi parameters for prediction the error is reduces and thus system has better accuracy. Real time checking and analysis also contributes to increase accuracy.

Communication methods of the systems are one more parameter that greatly differentiates these systems. Traditional system does not use any structured system and all manual checking of data which takes more time and thus there is no defined communication method for transmitting the data in emergency. Most of the existing IoT system works with internet connection. However for agricultural areas and remote rural areas, there are problems of net connectivity. In this system for data transfer the system uses both LoRa technology which is long range low power network and GSM module to transmit data in case there is no net work.

The traditional system used only warning to people through television channels or newspaper. Even the existing IoT based system have been designed to send alerts through email or mobile application which needs proper mobile net work connection for receiving them. However in the presented system due to using GSM modem it helps in sending real time SMS alerts even to basic mobile phone, so people in rural or farming areas do not need smartphone to get alerts. Real time SMS delivery provides more safety to people.

Cost is also a most critical factor to evaluate a system for implementation in real world conditions. Traditional system does not cost that much but it is unable to provide effective disaster management solution. Existing IoT based system has an additional component like server for analysis of data and for sending information which involves lots of cost and power consumption. The proposed system uses low cost components such as Raspberry pi and Arduino board with other simple sensors. Use of LoRa and GSM for data transmission reduces the overall cost by reducing reliance on cloud and the costly internet network which enables the system to be implemented economically across wide area.

Table 5.1 Comparison with Expected / Existing Outcomes

Feature	Traditional System	Existing IoT Systems	Proposed System
Monitoring	Manual	Automatic	Automatic
Prediction	No	Limited	Yes (ML-based)
Accuracy	Low	Medium	High
Communication	None	Internet-based	LoRa + GSM
Alert System	Delayed	Internet dependent	Real-time SMS
Cost	Low	High	Low

CHAPTER 6
CONCLUSIONS AND FUTURE SCOPE

6.1 Conclusions

The core objective of this project was to design and develop an IoT based flood prediction system for agricultural farms by using machine learning algorithms along with real-time environment monitoring. Based on the development, testing and evaluation which are carried out in previous chapters, it can be concluded that the aim of the project has been achieved. It integrates sensing, communication, data processing, prediction, visualization, and alerts in a single coherent system that is both technically feasible and practically viable.

One of the key achievements of this project is that a sensor node that collects real-time environment data has been successfully developed. It works with an Arduino microcontroller. It continuously monitors the amount of water present, rainfall, moisture in the soil, temperature and humidity that are critical for identifying floods. The continuous monitoring helps in sensing even small changes in the environment. The capability of a sensor node to collect data efficiently and effectively in real-time becomes a foundation of this system.

The successful use of the LoRa communication module is one of the major achievements. Since LoRa communication works with longer distances with less power, it becomes very practical for agriculture farms which are located in rural areas or having no network connectivity. The communication between the sensor node and the gateway works efficiently and has negligible data loss. It helps the system to work even in large areas without any proper network.

The data processing and analysis are done at the gateway node by a Raspberry Pi. The gateway node is used to gather data from LoRa module, to store it in a structured format and to run a machine learning algorithm to predict floods. Use of edge computing reduces the burden on the cloud.

Use of the Random Forest machine learning algorithm to predict floods in the farm is one of the prominent contributions. Instead of using a simple threshold based method which depends on fixed values of input parameters and thus lack accuracy, the machine learning model uses several input features to identify complex pattern from the input data and provides accurate flood predictions. During the testing, the model provided strong performance to distinguish between normal and flood conditions.

Visualization of the data and system performance through a web based interface as well as using the ThingSpeak platform can be called as a part of a successful achievement of this project. Such a interface allows the user to monitor and analyze the environment in the farm.

An SMS based alerting mechanism which can notify the user about the flood threat through GSM module is another achievement of this project. This mechanism proves very useful to rural people having lack of network access.

The designed system achieved all its objectives and clearly shows that it is possible to create an accurate, real-time and cost-effective flood prediction system using Internet of Things and machine learning technologies for agricultural applications. The project's success also paves the path for future improvements and wider implementation.

6.2 Limitations of the Work

Despite showing satisfactory accuracy, efficiency, and real-time capability, the developed IoT-based flood prediction system has several limitations found during development and testing:

1. Limited Sensor Accuracy and Reliability

The system is entirely reliant on sensor data for making predictions. Thus the reliability and accuracy of the sensor itself has a direct impact on the output of the machine learning model. In a real-world scenario, the sensors may provide misleading readings in response to external factors such as environmental noise, changes in temperature and humidity, and electromagnetic interference. Poor calibration and inadequate maintenance can result in erratic data, which in turn will affect the system’s ability to correctly classify states as either flood or not flood. For example, a slight variation in the water level and soil moisture values will directly alter the predicted state of a flood. The usage of appropriate calibration techniques and regular monitoring/maintenance of the sensors is necessary.

2. Insufficient Data size

Despite good test performance, the machine learning model used is based on a small dataset that was collected over a period of 1.5 months. Although the algorithm has recognized certain patterns, better performance is expected with more comprehensive data. The limited size may have failed to cover different seasonal variations and environmental influences, therefore, not being able to fully generalize for future occurrences. This could be improved with collection of data from different regions, seasons and diverse environmental settings.

3. Unrealistic Testing Environment

The test environments were limited to the controlled laboratory or outdoor environment; this is somewhat different to a real world agricultural farm. Extreme weather conditions, hardware damage due to environmental factors and the general impact on hardware due to long term exposure can reduce overall system performance. The system may also develop faults such as circuit failures and sensor degradations that will effect overall function of the system over a period of time.

4. Lack of Multi-level Classification

The current ML algorithm used is capable of binary classification only, distinguishing the state between "flood" or "not flood". Though it can provide notification of potential floods, the model lacks a multi-level classification feature that could predict how high the water level would be, such as low risk, medium risk and high risk. This can cause users to make less informed decisions and actions to tackle floods.

5. GSM Network Dependability

The alert notification through SMS is dependent on a reliable GSM network. While a GSM module is capable of providing alerts where internet connectivity is unavailable, weak or absent network coverage at the testing sites can impact the system. This issue can render the system useless where its operation is the most critical.

6. Power Constraints

The system has to run on batteries since it is designed for the purpose of being placed in remote farms. Continuous power is essential for reliable operation of all components, but the supply can be intermittent at times. This may cause the system to have temporary breakdowns that could lead to the loss of communication. Low power consuming modules such as LoRa are utilized in order to reduce the system's dependency on a strong power source.

7. Scalability and Complexity of the System

The proposed system can be implemented over a vast region but the scale-up needs to be designed and planned properly due to factors like placement and configuration of multiple nodes, maintenance and monitoring across all the sites and system reliability at each location.

8. Narrow Scope of Application

The system has been designed for flood prediction only for agricultural farms. There might be a requirement of some additional functionality, such as integration with other weather prediction systems and use of satellite data or some advanced predictive models considering the long-term climate changes in future and in turn improve the precision.

These limitations help in providing an outlook for improvement and future research in order to make this system even more efficient and feasible in the real world applications.

Table 6.1 – Current Limitations

Limitation	Description
Dataset Size	Dataset limited to prototype testing period; seasonal variation not captured
Single Node	Only one sensor node deployed; multi-node coverage not demonstrated
Power Supply	Mains-powered prototype; battery/solar operation not implemented
Outdoor Enclosure	IP-rated weatherproof enclosure not fabricated for the prototype
Model Retraining	Model requires manual retraining; no automated online learning
GSM Coverage	Dependent on GSM network availability in field location

6.3 Future Scope

The envisioned IoT based flood prediction system represents a solid basis for the future expansion of both Smart Agriculture and disaster management research. Although the presented system proves the viability of a real time flood monitoring and prediction system, several enhancements could significantly increase the system's efficiency, accuracy and practicality in real world applications:

• One area where more advanced Machine Learning and Deep learning techniques can be incorporated. The Random forest algorithm used in this system has been effective in classification, however, it may be possible to integrate and research the use of more advanced models like LSTMs. LSTMs are particularly beneficial in dealing with time-series data, they are able to recognize temporal dependent relationships in data and learn to predict future trends based on past data. As parameters like rainfall and soil moisture vary through time, it is highly possible that incorporating LSTMs and other recurrent neural network architectures may improve flood predictions:

• More sensor and data source integration. Although current system utilizes parameters like water level, soil moisture, temperature and humidity, more sensor inputs may be used. Water flow and pressure sensors could give extra insight in underground water levels and water flow. External data sources can also be integrated like real-time weather predictions by interfacing through APIs which could provide input into the algorithm for potential future rain events.

• A dedicate mobile/web application could be designed. Current application includes a web based display of data and alert messages delivered via SMS, a full mobile/web application would provide a much more intuitive and engaging user experience, incorporating real time display and graphical data presentation, interactive graphs of stored data, a more detailed user interface, and also control of the whole system remotely.

• Integration of GPS and map based technology. If each sensor node had a GPS capability, all the data can be put on a map, and it will be easier to plot flooded areas and their size. This will help people to navigate away from or towards affected areas based on their needs, for instance, on their way to an evacuation centre or taking their livestock to a safety.

• Utilization of renewable energy. All sensors and components of the system should ideally run off some renewable energy source like solar power, as sensors are likely to be situated in rural and distant locations where an AC power source might not be available, therefore increasing reliability and sustainability of the system.

• Interfacing with a government disaster management system. All data from many sensor nodes would need to be sent to the relevant agencies for their interpretation and decision making. Through linking up to government agencies the information can be collated into regional or even national flood alerts.

• Implementing a multi-level risk classification. In this thesis the system classifies situations into two categories; flooded or not flooded. In future iterations, the system should be able to produce multiple levels of classification e.g. Low Risk, Moderate risk and High Risk. This would allow users to know when action needs to be taken based on the level of severity of the flood.

• Improving system scalability. As the demand for the smart farming and flood prediction increases, the system should be scalable to cope with a greater number of sensor nodes spread across larger areas.

• Increasing the robustness and durability of the system. All components of the system are exposed to the elements and as such they need to be highly durable. This can include research into weather resistant casing, sensor lifetime improvement and more resilient network design.

Table 6.2 – Proposed Future Enhancements

Enhancement	Description	Priority
Solar Power	Add solar panel + Li-ion battery for field deployment	High
Multi-node Network	Deploy multiple sensor nodes with a single Raspberry Pi gateway	High
LSTM Model	Replace Random Forest with LSTM for temporal pattern learning	Medium
Mobile App	Develop a mobile application for real-time alerts and maps	Medium
GPS Tagging	Add GPS module to geo-tag sensor readings	Medium
Automated Retraining	Schedule periodic model retraining as dataset grows	Medium
LoRaWAN Migration	Upgrade to LoRaWAN for multi-gateway scalability	Low
Government API Integration	Connect to national flood forecasting portals	Low

SOURCE CODE

1. dashboard.py

Code:
```
from flask import Flask, render_template
import pandas as pd
app = Flask(_name_)
@app.route("/")
def home():
    data = pd.read_csv("flood_data.csv")
    latest = data.tail(10)
    return latest.to_html()
app.run(host="0.0.0.0", port=5000)
```

2. Lora_Server.py

Code:
```
from SX127x.LoRa import *
from SX127x.board_config import BOARD
import time
import requests
import csv
import os
import joblib
import numpy as np
# --------- ThingSpeak ---------
THINGSPEAK_API = "T30N64145WCUBZTC"
# --------- CSV file ---------
CSV_FILE = "flood_data.csv"
# --------- Load ML Model ---------
model = joblib.load("flood_model.pkl")
# Create file if not exists
```

```
if not os.path.exists(CSV_FILE):
    with open(CSV_FILE, mode="w", newline="") as f:
        writer = csv.writer(f)
        writer.writerow(["Time", "Temperature", "Humidity", "Water_Level",
"Soil", "Flood"])
BOARD.setup()
class LoRaReceiver(LoRa):
    def __init__(self):
        super(LoRaReceiver, self).__init__()

        self.set_mode(MODE.SLEEP)
        self.set_freq(433.0)
        self.set_bw(BW.BW125)
        self.set_spreading_factor(7)
        self.set_coding_rate(CODING_RATE.CR4_5)
        self.set_preamble(8)
        self.set_sync_word(0x34)
        self.set_dio_mapping([0] * 6)
    def start(self):
        print("Receiver Started")
        self.set_mode(MODE.RXCONT)
        while True:
            flags = self.get_irq_flags()
            if flags['rx_done']:
                payload = self.read_payload(nocheck=True)
                data = bytes(payload).decode()
                print("Received:", data)
```

```
            try:
                # Expected format: temp,hum,level,soil
                temp, hum, level, soil = map(float, data.split(","))
                # -------- ML Prediction --------
                pred = model.predict(np.array([[temp, hum, level, soil]]))[0]
                print("Flood Prediction:", pred)
                # -------- Upload to ThingSpeak --------
                url                                                    =
f"https://api.thingspeak.com/update?api_key={THINGSPEAK_API}&field1={temp}
&field2={hum}&field3={level}&field4={soil}&field5={pred}"
                requests.get(url)
                print("Uploaded to ThingSpeak")
                # -------- Save in CSV --------
                with open(CSV_FILE, mode="a", newline="") as f:
                    writer = csv.writer(f)
                    writer.writerow([
                        time.strftime("%Y-%m-%d %H:%M:%S"),
                        temp, hum, level, soil, pred ])
                print("Saved in CSV")
            except Exception as e:
                print("Error:", e)
            self.clear_irq_flags(RxDone=1)
        time.sleep(0.5)
lora = LoRaReceiver()
lora.start()
```

3. **Lora_Server_SMS.py**

Code: from SX127x.LoRa import *

from SX127x.board_config import BOARD

```
import time, csv, requests, joblib, numpy as np
import serial
import RPi.GPIO as GPIO
GPIO.setwarnings(False)
# -------- Load ML Model --------
model = joblib.load("flood_model.pkl")
# -------- ThingSpeak --------
THINGSPEAK_API = "T30N64145WCUBZTC"
# -------- GSM --------
PHONE = "+919347574356"
def send_sms(msg):
    gsm = serial.Serial("/dev/ttyUSB0", 9600, timeout=1)
    time.sleep(2)
    gsm.write(b'AT+CMGF=1\r')
    time.sleep(1)
    gsm.write(f'AT+CMGS="{PHONE}"\r'.encode())
    time.sleep(1)
    gsm.write(msg.encode())
    gsm.write(b"\x1A")
    time.sleep(3)
    print("SMS Sent")
BOARD.setup()
class LoRaReceiver(LoRa):
    def __init__(self):
        super().__init__()
        self.set_mode(MODE.SLEEP)
        self.set_freq(433.0)
```

```
        self.set_bw(BW.BW125)
        self.set_spreading_factor(7)
        self.set_sync_word(0x34)
    def start(self):
        print("System Ready")
        self.set_mode(MODE.RXCONT)
        while True:
            flags = self.get_irq_flags()
            if flags['rx_done']:
                data = bytes(self.read_payload(nocheck=True)).decode()
                print("Received:", data)
                try:
                    temp, hum, level, soil = map(float, data.split(","))
                    # ML prediction
                    pred = model.predict(np.array([[temp, hum, level, soil]]))[0]
                    print("Flood:", pred)

                    # Upload to ThingSpeak
                    url = f"https://api.thingspeak.com/update?api_key={THINGSPEAK_API}&field1={temp}&field2={hum}&field3={level}&field4={soil}&field5={pred}"
                    requests.get(url)
                    print("Uploaded to ThingSpeak")
                    # GSM Alert
                    if pred ==1:
                        send_sms("FLOOD ALERT! Water level high.")
                except:
                    print("Format error")
```

self.clear_irq_flags(RxDone=1)

time.sleep(0.5)

LoRaReceiver().start()

4. train_model.py

Code: import pandas as pd

from sklearn.ensemble import RandomForestclassifier import joblib

data = pd.read_csv("flood_data.csv")

X = data [['Temperature', 'Humidity', 'Water_Level', 'Soil']]

y = data['Flood']

model = RandomForestclassifier() model.fit(X, y)

joblib.dump(model, "flood_model.pkl") print("Model ready")

5. test_accuracy.py

Code: import pandas as pd

from sklearn.ensemble import RandomForestClassifier

from sklearn.model_selection import train_test_split

from sklearn.metrics import accuracy_score, precision_score, f1_score

import joblib

Load data

data = pd.read_csv("flood_data.csv")

❗ REMOVE EMPTY FLOOD VALUES

data = data.dropna(subset=['Flood'])

Features

X = data[['Temperature', 'Humidity', 'Water_Level', 'Soil']]

Target

y = data['Flood']

Split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

```python
# Train
model = RandomForestClassifier()
model.fit(X_train, y_train)
# Predict
y_pred = model.predict(X_test)
# Metrics
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred, zero_division=1))
print("F1 Score:", f1_score(y_test, y_pred, zero_division=1))
# Save
joblib.dump(model, "flood_model.pkl")
print("Model saved")
```

6. Arduino_TX_code.cpp

```cpp
Code:#include <SPI.h>
#include <LoRa.h>
#include <DHT.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
// --------- LCD ---------
LiquidCrystal_I2C lcd(0x27, 16, 2);
// --------- DHT11 ---------
#define DHTPIN 4
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
// --------- Ultrasonic ---------
#define TRIG_PIN 5
#define ECHO_PIN 6
```

```
// --------- Soil Sensor ---------
#define SOIL_PIN A0
// --------- LoRa Pins ---------
#define SS 10
#define RST 9
#define DIO0 2
void setup() {
  Serial.begin(9600);
  // LCD setup
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("Flood Monitor");
  delay(2000);
  lcd.clear();
  // DHT setup
  dht.begin();

  // Ultrasonic
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  // LoRa setup
  LoRa.setPins(SS, RST, DIO0);
  if (!LoRa.begin(433E6)) {
    lcd.print("LoRa Failed!");
    while (1);
  }
```

```
  lcd.print("LoRa Ready");
  delay(2000);
  lcd.clear();
}
// --------- Ultrasonic ---------
long readWaterLevel() {
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);
  long duration = pulseIn(ECHO_PIN, HIGH);
  long distance = duration * 0.034 / 2;
  return distance;
}
void loop() {
  float temp = dht.readTemperature();
  float hum = dht.readHumidity();
  long water = readWaterLevel();
  int soil = analogRead(SOIL_PIN);
  // --------- Serial ---------
  Serial.print("T:"); Serial.print(temp);
  Serial.print(" H:"); Serial.print(hum);
  Serial.print(" W:"); Serial.print(water);
  Serial.print(" S:"); Serial.println(soil);
  // --------- LCD Display ---------
  lcd.clear();
```

```
    lcd.setCursor(0, 0);
    lcd.print("T:");
    lcd.print(temp);
    lcd.print(" H:");
    lcd.print(hum);
    lcd.setCursor(0, 1);
    lcd.print("W:");
    lcd.print(water);
    lcd.print(" S:");
    lcd.print(soil);
    // --------- LoRa Send ---------
    LoRa.beginPacket();
    LoRa.print(temp);
    LoRa.print(",");
    LoRa.print(hum);
    LoRa.print(",");
    LoRa.print(water);
    LoRa.print(",");
    LoRa.print(soil);
    LoRa.endPacket();
    Serial.println("Data Sent");
    delay(3000);
}
```

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APPENDICES

Appendix A: Contributions by Team Members

The project titled *"IoT Based Flood Predictive System for Agriculture Fields Using Machine Learning"* was successfully carried out as a collaborative effort by a dedicated team of students under the guidance and supervision of Dr. Malleti Sreedhar, Professor in the Department of Electronics and Instrumentation Engineering at VNR Vignana Jyothi Institute of Engineering and Technology, Hyderabad. The successful completion of this project was made possible through the coordinated contributions of each team member, who played specific roles across various phases including system design, hardware development, software implementation, data processing, testing, and documentation. Each member brought unique skills and expertise, ensuring that the project was executed efficiently and met its intended objectives.

Gunji Dhana Laxmi (Registration Number: 22071A1018) played a crucial role in the hardware development aspects of the project, particularly focusing on the transmitter and gateway nodes. She was responsible for interfacing multiple sensors, including water level sensors, rainfall sensors, soil moisture sensors, and temperature and humidity sensors, with the Arduino microcontroller. Her work involved careful wiring, circuit assembly, and ensuring proper communication between components. She also handled the testing and calibration of sensors to ensure accurate data collection under different environmental conditions. Additionally, she worked extensively on integrating and testing communication modules such as LoRa and GSM, ensuring reliable wireless data transmission and alert functionality. Her contribution also included the preparation of detailed circuit diagrams, which were essential for both system understanding and documentation purposes.

Jejjari Vinay (Registration Number: 22071A1021) was primarily responsible for the overall system design and integration of both hardware and software components. His role was central to ensuring that all parts of the system worked seamlessly together. He implemented the machine learning component of the project by deploying a Random Forest model on the Raspberry Pi, which was used to analyze environmental data and predict flood conditions. He also developed Python scripts for data logging, real-time monitoring, and automation of alerts through SMS notifications. In addition, he contributed to building the system dashboard for visualization and monitoring of sensor data. His responsibilities extended to coordinating experimental trials, validating system performance, and troubleshooting integration issues. His leadership in system integration ensured that the project achieved its functional goals and operated reliably in real-time scenarios.

Tokala Vaishnavi (Registration Number: 22071A1058) contributed significantly to the software support, data handling, and documentation aspects of the project. She was responsible for creating and managing the ThingSpeak channel used for cloud-based data visualization and monitoring. In addition, she worked on developing a local web-based dashboard to provide an alternative interface for viewing system data. Her role in dataset preparation and preprocessing was critical for the effective training and evaluation of the machine learning model. She assisted in model evaluation by analyzing performance metrics and ensuring the accuracy of predictions. Furthermore, she played a key role in documenting the project methodology, results, and observations in a structured and professional manner. Her contributions also included the preparation of figures, tables, and graphical representations that enhanced the clarity of the report and presentation. She actively supported the drafting of the final report and presentation materials, ensuring that the project outcomes were communicated effectively.

Overall, the project reflects a strong example of teamwork, where each member contributed meaningfully to different aspects of the system. The integration of hardware, software, and machine learning components required continuous collaboration and coordination among team members. The division of responsibilities allowed for efficient execution while also enabling knowledge sharing and skill development within the team. Under the guidance of Dr. Malleti Sreedhar, the team was able to successfully design, develop, and implement a functional IoT-based flood prediction system tailored for agricultural applications. The collective efforts of the team members ensured that the project not only met academic requirements but also demonstrated practical relevance and potential for real-world deployment.

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APPENDIX B: RELEVANCE TO SUSTAINABLE DEVELOPMENT GOALS(SDGs)

The suggested Flood Prediction System for Agriculture Fields using Machine Learning is very much related to a number of SDGs because it helps in solving environmental issues, increasing agricultural sustainability, and making sure agriculture is practiced sustainably.

SDG 2: Zero Hunger
The present study supports this goal as it will help farmers predict flooding, which might damage their crops. If they know about any upcoming flood, then farmers can take preventative actions like changing their irrigation techniques or even harvesting their crops early.

SDG 6: Clean Water and Sanitation
Prediction of floods becomes essential when it comes to managing water resources efficiently. The project ensures that farmers know how the water flows and whether there is excessive water in their fields. This could have negative implications on the soil fertility and crop yield.

SDG 9: Industry, Innovation, and Infrastructure
Machine learning, sensors, and cloud computing employed in this project will enhance innovation in the agricultural sector. The innovation enhances resilient infrastructure development through the introduction of smart monitoring technology that can be applied in other environmental monitoring purposes.

SDG 11: Sustainable Cities and Communities
Floods affect not only agricultural fields but also surrounding urban and rural areas. The ability of the system to offer early warnings is critical in enhancing safety and reducing risks to these communities

SDG 12: Responsible Consumption and Production
The ability to minimize losses caused by floods will enable farmers to use agricultural resources such as water, fertilizer, and energy in the most effective manner possible.

Goal 13: Climate Action
Climate action plays a significant role in this project, since it tries to deal with the increasing occurrence of extreme weather conditions, including floods, caused by climate change. The system increases the ability to adapt as it allows using information for decision-making, and gives an opportunity to predict and reduce risks.

Goal 15: Life on Land
Soil erosion may be the consequence of floods. Thus, by preventing this phenomenon, one contributes to protecting soils.
Thus, the project contributes to sustainable agriculture, disaster risk reduction, and climate adaptation, applying modern technologies. It will help to benefit the environment and promote sustainability.

Project demonstration at SHOW AND TELL